Nord*Lit*

Manual

Nordic Literature Search

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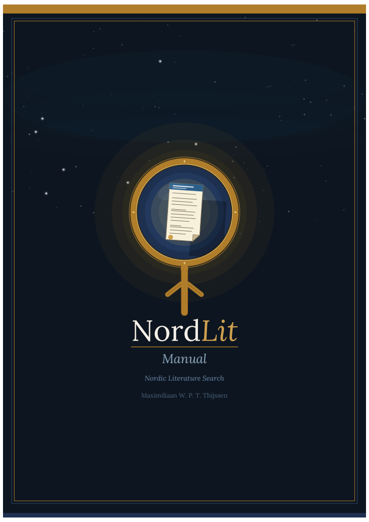

Nord*Lit*

The mark shows a magnifying glass, the universal symbol of search, whose lens frames an open document page, representing the literature being sought. The handle is cast in the shape of the Elder Futhark rune *Tiwaz*, an ancient Nordic symbol of guidance and direction. Together they express the purpose of Nord*Lit*: to uncover Nordic scholarship and point researchers toward knowledge.

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> Bugs, issues, and suggestions for improvement are very welcome. Please send them by e-mail to the address above — all feedback is read and appreciated.

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1 Why NordLit

NordLit exists to make Nordic research easier to find, use, and document. Across the Nordic countries, a large amount of valuable research is available through university repositories, national research portals, library catalogues, and local journal platforms. Much of this material is well maintained and openly available. Yet it is not always captured consistently in the international databases that many users begin with.

This creates a practical problem. A search can appear broad and systematic while still missing relevant Nordic research simply because that research is distributed across many national and institutional systems. In other words, the issue is often not that the research is unavailable, but that it is harder to discover in a consistent way.

NordLit is designed to address that problem. NordLit does not replace the national or institutional services that already collect, describe, and host research outputs. Those services remain the authoritative sources for their own records. Instead, NordLit provides a shared search layer across Nordic research metadata. It brings together descriptive information such as titles, authors, abstracts, publication years, identifiers, and stable source links, so that users can search across countries in one place and then follow each record back to its original source.

This approach matters because many important research questions do not stop at national borders. People often need Nordic literature when preparing reports, evidence reviews, policy briefs, research projects, or comparative studies. In fields such as education, welfare, labour markets, health, and public administration, the Nordic countries are frequently studied together. They are not identical, but they are similar enough in important institutional ways that comparison is often meaningful. When the question is regional, the search must also be regional.

At present, that kind of regional search is often more difficult than it should be. Relevant publications are spread across multiple services, each built for its own national or institutional context. These services may all work well on their own terms, but they often differ in how they handle search, filtering, and export. This forces users to repeat the same work several times: running similar searches in multiple places, adapting terms to different systems, and then combining the results manually. This is inefficient, but it also reduces transparency and increases the likelihood of errors. When a search must be rewritten for each platform, it becomes harder to document clearly and harder for another person to repeat.

NordLit is built on a simple principle: if (reliable) records of Nordic research already exist, users should be able to search them together without rebuilding the same search strategy all over again. By connecting infrastructures through a common discovery layer, NordLit reduces duplicated effort, improves the visibility of Nordic research, and supports more transparent documentation of how evidence was gathered. At the same time, it respects the boundaries that matter. It does not take over the role of national services, and it does not claim ownership of the material it points to. Its purpose is to help users find what already exists, understand where it comes from, and use it with greater confidence.

1.1 Who NordLit Is For

NordLit is intended for those who need to find Nordic research in a systematic and understandable way. This includes researchers, librarians, students, analysts, policy staff, and others working with evidence, literature reviews, mapping studies, or background reports.

It is especially useful when the task requires coverage across more than one Nordic country. In such cases, users often need a search process that is not only broad, but also consistent and explainable. NordLit is designed to support that need.

1.2 What Goes Wrong Now

Each Nordic country has invested in infrastructures to register and disseminate research outputs. These systems often work well within their own national settings. The problem arises when a user needs to search across the region as a whole.

In practice, cross-Nordic searching often means moving between several national platforms, learning how each one works, and adjusting the search to fit different rules. Even when two services seem similar, they may handle phrases, truncation, dates, names, or subject fields in different ways. Small differences in search behaviour can change what is retrieved.

This fragmentation is not only inconvenient. It also increases the risk that relevant material is missed. That matters in any serious review of the literature, but it becomes especially important in work that depends on transparency and completeness, such as systematic reviews or evidence syntheses. If the same search must be rewritten repeatedly across different interfaces, it becomes harder to explain exactly what was done, harder for someone else to repeat it, and easier for small inconsistencies to affect the final result.

A simple example illustrates the point. Imagine a reviewer searching for studies on school attendance interventions in Nordic contexts. A careful strategy might combine concepts such as attendance, intervention, school, outcomes, and Nordic-language equivalents. If one platform supports phrase searching, another handles phrases differently, and a third offers only limited control over how terms are combined, then the search can no longer be applied in the same way everywhere. At that point, it becomes difficult to know whether missing studies are truly absent or whether they were missed because the search process could not be implemented consistently.

1.3 Why Nordic Coverage Matters

International databases are important and useful, but they do not provide a complete picture of Nordic research. This is largely a structural issue. Many Nordic research outputs are published or archived through national and institutional channels, including repositories, local journals, library-based platforms, and report series. Some are published in Nordic languages. Others appear in formats that are highly relevant in national or regional contexts but less visible in large international indexes.

As a result, searches that rely only on global databases can underrepresent Nordic scholarship. This is not because the work lacks value or relevance, but because its route to visibility is different. A regional search service helps correct that imbalance by improving the chances that Nordic research is found when it should be included.

This matters both for recall and for fairness. Better recall means finding more of the relevant literature. Fairness means giving Nordic research proper visibility in fields where Nordic evidence is directly relevant to the question being asked.

1.4 A Shared Search Layer

NordLit is best understood as a shared discovery layer for Nordic research metadata. It does not replace the systems that hold the original records. Instead, it allows users to search across them in a more unified way.

The focus is on bibliographic metadata, that is, descriptive information about research outputs. This includes fields such as title, author, abstract, publication year, publication type, identifiers such as DOI where available, and stable links back to the source record. The full text remains where it already belongs: with the institution, repository, journal platform, or publisher responsible for hosting it.

This design has several important advantages. It respects that national and institutional systems are authoritative for their own records and are responsible for how those records are maintained. It also makes NordLit more manageable in legal and practical terms. Working with metadata is usually less sensitive than collecting and redistributing full texts, and it allows each source to remain in control of access conditions, licences, corrections, and takedown procedures.

For users, this means that NordLit boosts discoverability without breaking provenance. A result in NordLit should not appear detached from its origin. It should remain clear where the record comes from and how to reach the original source.

1.5 What Metadata Means

The term metadata may sound technical, but the basic idea is simple. Metadata is the information that describes a publication and helps people identify and find it again. In a library catalogue, the catalogue record is metadata. In a citation manager, the fields for title, author, year, and journal are metadata. In a repository or journal portal, the descriptive page about a publication is largely metadata.

NordLit works with this descriptive layer. That is what makes it possible to search records in a more consistent way, display them in a common format, and export them for further use. It also helps clarify what NordLit is not. It is not a publisher, and it does not claim ownership of the works it indexes. Its role is to help users locate and understand research outputs, not to replace the source that holds them.

1.6 What Users Can Do in NordLit

At its core, NordLit helps users do three things: search, inspect, and export.

Firstly, users can search across Nordic records in one place rather than repeating the same process in several national systems. Secondly, they can inspect the descriptive information attached to each result, including where the record comes from and how to follow it back to the original source. Thirdly, they can export records and search results in formats that support citation management, screening workflows, documentation, and later analysis.

These functions are important not only for convenience, but also for clarity. A useful search service should not merely retrieve records. It should also make it easier to understand what was found, where it came from, and how the results can be used responsibly afterwards.

1.7 Search You Can Repeat

NordLit is being developed with transparent searching in mind. In many kinds of research work, especially evidence synthesis, finding relevant studies is only part of the task. It is also necessary to explain how the studies were found. This usually means documenting search strings, dates of search, filters, sources, and export steps.

A shared discovery layer supports this kind of work. Instead of translating a search across several different interfaces, a user can apply one documented search strategy to a harmonised index. This makes the process easier to describe and easier to repeat. It also makes collaboration easier, because other users can rerun the same search in the same environment rather than depending on several different systems with different rules.

This does not mean that all uncertainty disappears. Records can change over time when repositories add abstracts, correct errors, update links, or improve descriptions. That is normal and often beneficial. The aim is not to freeze the data, but to make change easier to manage by supporting consistent search behaviour, stable exports, and clear source attribution.

1.8 Transparency in Practice

Transparency is one of NordLit's central goals. In practical terms, this means that users should be able to see what a record is, where it comes from, and how it connects back to the original source. A search is more trustworthy when others can understand it, inspect it, and repeat it.

Transparency also means being clear about what NordLit can and cannot do. NordLit can improve discovery, support documentation, and reduce duplicated work. It cannot guarantee perfect completeness, and it cannot remove every difference that exists between underlying data sources. What it can do is make those differences more visible and more manageable.

For non-technical users, this is especially important. A system should not feel like a black box. The aim is for users to understand not only what NordLit returns, but also why a result appears and how it relates to the original record.

1.9 Why Not BASE-Search?

Users may reasonably ask why an existing academic search service such as BASE (Bielefield Academic Search Engine) is not sufficient. BASE is useful in many contexts and offers broad coverage through a standardised search interface. However, it does not fully solve the practical problems that NordLit is intended to address.

One issue is export. For users working with literature reviews, evidence synthesis, or other forms of structured searching, export options are not a minor feature. They are part of the research workflow. If export is limited, the user may still face substantial manual work after the search itself. NordLit places greater emphasis on export because the ability to take results out of the system clearly and efficiently is essential for transparent research practice.

Another issue is speed and responsiveness. A service can be technically functional while still feeling difficult to use if it is slow at important stages of the workflow. When users need to test, refine, and compare searches, delays make the process harder and less predictable.

Filtering is also important. If users cannot combine several filters at once, they have less control over how they narrow a result set. This can make searching less flexible and less precise, especially when dealing with large numbers of records.

There is also a question of clarity in how results are matched. A system that searches keywords from unclear or inconsistently explained sources may be useful for broad discovery, but it can be harder to understand and document in a transparent way. For users who need to explain their search methods carefully, it matters whether the searchable fields are clearly defined.

Finally, even with a standardised interface, users may still need to search separately by data provider. This means that the underlying fragmentation has not really disappeared. It has only been presented through a common access point. NordLit is intended to go further by supporting a genuinely shared Nordic discovery workflow rather than requiring users to reproduce provider-by-provider searching through a single external platform.

For these reasons, the question is not whether BASE is useful in general. It is. The question is whether it is designed for the specific problem NordLit is trying to solve. NordLit is being developed because cross-Nordic searching requires strong export options, clearer control over search behaviour, better support for documentation, and a workflow built around regional discovery rather than general aggregation alone.

1.10 What NordLit Does Not Do

NordLit has a clear purpose, but that purpose also has limits. It does not replace national repositories, library catalogues, or journal platforms. It does not decide what counts as good research, and it does not function as a final authority on any record.

It also does not host or republish full texts. Its role is to improve discovery, not to take over distribution. The original source remains the place to verify the record in full, check access conditions, and consult the publication itself.

Being clear about these limits is part of being transparent. A service is easier to trust when its role is defined honestly and without exaggeration.

1.11 Limits and Trade-offs

A shared Nordic discovery layer has clear benefits. It reduces duplicated work, improves the visibility of regionally relevant research, and makes cross-border searching easier to document and repeat. It can also give smaller publication channels more visibility by placing them within a broader discovery context.

At the same time, there are unavoidable limits. A metadata-based service depends on the quality of the data it receives. Some repositories provide richer abstracts than others. Some records, especially older ones, may be incomplete. The same publication may appear in several places with slightly different descriptions, making deduplication difficult. NordLit can reduce these problems through careful matching and transparent source attribution, but it cannot eliminate the underlying variation altogether.

It is therefore best to describe NordLit as a pragmatic infrastructure. It is not a promise of perfect completeness, and it is not an attempt to erase differences between Nordic publishing systems. It is a practical way to make existing research outputs easier to find and use responsibly.

1.12 Lawful and Responsible Use

NordLit is designed to operate within existing legal and ethical frameworks. It focuses on metadata that is available for reuse under the relevant conditions. It does not collect, store, or redistribute full texts as part of its core function. Where rights statements or licence information are available, these should be carried through and presented clearly.

Author names, affiliations, and related descriptive information are part of normal scholarly communication. Even so, NordLit treats such information with care and for a clear purpose: discovery, attribution, and traceability. The goal is not profiling or secondary exploitation, but helping users identify research outputs correctly and connect them to their original sources.

Responsible use also means being honest about uncertainty. Not every record will be equally complete, not every source will describe publications in the same way, and not every search will retrieve everything relevant on the first attempt. A trustworthy system acknowledges these realities and helps users work with them openly.

1.13 What Users Should Expect

Users should expect NordLit to make Nordic research easier to find, easier to compare across countries, and easier to document. They should expect clear source links, understandable record information, and export options that support further work.

They should also understand what NordLit is not. It is a discovery service, not the final authority on a publication. The original source remains essential for verification, access, and full context. NordLit adds value by making the route to those sources simpler, clearer, and more consistent across the Nordic region.

2 National Sources

NordLit builds on the fact that every Nordic country already has services that register, describe, or disseminate research outputs. The problem is usually not that the material is missing. The problem is that it is spread across several systems designed for national, institutional, or library-specific use. NordLit therefore does not try to replace these services. Instead, it connects to them through a shared discovery layer so that users can search across borders more consistently while still following each record back to the original source.

It is also important to be realistic about change. National infrastructures evolve. Services are redesigned, merged, or expanded; new institutions join; and technical interfaces are improved over time. For that reason, NordLit gives priority to sources that provide stable metadata, clear provenance, and dependable links back to an authoritative record. This means favouring maintained infrastructures over one-off exports or fragile approaches such as website scraping.

In total, NordLit currently draws from roughly seventy-five distinct sources. These range from very large national aggregators containing millions of records to individual journals that publish a few dozen articles per year. Each source plays a specific role in the wider Nordic research landscape, and together they cover the ground that no single source could cover on its own.

2.1 Norway

2.1.1 NVA

What it is. NVA, or *Nasjonalt vitenarkiv*, is Norway's national infrastructure for registering and making research outputs available through a shared service. Sikt describes NVA as a common national solution that collects publications and other research results and makes information about Norwegian research available in one place. It also brings together functions that were previously divided between Cristin and local institutional repositories. In practice, this means that publications and related research information can be surfaced through one national service rather than through many unrelated local entry points.

Why NordLit uses it. For NordLit, NVA matters because it offers a national route into Norwegian research metadata that is substantially more coherent than harvesting institution by institution. Without such an infrastructure, Norwegian material would need to be discovered through many separate repositories and local systems, each with its own metadata conventions, repository practices, and levels of detail. NVA reduces this fragmentation by exposing records through a shared national framework. This does not make all metadata identical, but it does create a more stable and predictable discovery environment.

This is especially important for a cross-border service such as NordLit. Users searching across the Nordic region benefit not only from broad coverage, but also from clearer provenance and more consistent institutional context. A record in NVA is more readily understood as part of a national infrastructure, and it is often easier to trace the responsible institution, researcher affiliation, and landing page than it would be in a purely local system. In that sense, NVA is valuable not simply because it contains Norwegian records, but because it helps make Norwegian research legible at scale.

A concrete example is the difference between searching for Norwegian research through scattered institutional archives and searching through a national aggregation point like NVA. In the first case, NordLit would need to identify relevant repositories one by one, harmonise their descriptive information separately, and handle inconsistent links and institutional naming. In the second, NVA provides a shared route that can simplify integration, improve consistency in presentation, and preserve a clearer line back to the institution responsible for the record. That makes NVA strategically important even when the underlying metadata remains uneven.

What you can expect to find. Users should expect NVA to contain a broad range of Norwegian research outputs and related research information. In practice, this means journal articles, books and book chapters, reports, conference contributions, artistic and dissemination outputs, doctoral dissertations, master's theses, and other registered research results. Records are typically linked to researcher profiles, projects, and institutional affiliations. They commonly include a stable landing page and may also include persistent identifiers such as DOIs, ORCID links, institutional names, publication channels, and licence information where available. At the time this manual was written, NVA's coverage in NordLit numbers in the low millions of records, giving it by a wide margin the largest presence among Norwegian sources.

For NordLit users, one of the most important consequences is that NVA can reveal material that may be less visible in international indexes. Large international databases often privilege certain publication types, publishers, or language areas, whereas a national infrastructure is better placed to expose institutionally registered outputs, Norwegian-language scholarship, and material with stronger local or regional relevance. For example, a report produced within a Norwegian research institute, or a publication deposited through a university repository workflow, may be easier to identify through NVA than through a commercial citation index.

At the same time, users should expect variation. Some records are rich and immediately useful for discovery: they may include an abstract, keywords, multiple identifiers, ORCID links, full institutional metadata, and direct access to an open version of the work. Other records are thinner: they may provide only the bibliographic basics and function mainly as signposts pointing the user onward to the source institution, publisher page, or repository copy. From a discovery perspective, both kinds of records are useful, but they support different levels of downstream reuse. Rich records support filtering, linking, and interoperability more effectively; minimal records still help establish that the work exists and where responsibility for it lies.

Advantages and limitations. The main advantage of NVA is coherence at the national level. It gives NordLit a practical way to include Norwegian material without treating every university, college, museum, or research institute as a completely separate discovery problem. This national coherence has several benefits. First, it lowers technical and editorial complexity for integration. Second, it improves the visibility of provenance, because records remain connected to the institutions responsible for registering and maintaining them. Third, it supports more even discovery across the Norwegian research landscape, including material that might otherwise remain locally visible but internationally obscure.

Another advantage is that NVA is not merely a search layer; it is part of a wider national infrastructure for research information management. Sikt explicitly connects NVA to registration, sharing, reuse, and reporting workflows. That matters because metadata created within a system that also supports institutional and national reporting has stronger incentives for maintenance, verification, and standardised use than metadata exposed only for ad hoc local discovery. For NordLit, this increases the likelihood that NVA records will remain stable, institutionally anchored, and meaningful over time.

The main limitation, however, is that a common service does not eliminate differences in source quality. NVA can gather and expose records through a shared infrastructure, but it cannot fully standardise the descriptive practices of all contributing institutions or create information that was never recorded in the first place. Metadata richness may therefore differ substantially between records. Some institutions may contribute detailed records with persistent identifiers and rich contextual information, while others may contribute more limited descriptions. As a result, NordLit can use NVA to improve access, discoverability, and consistency of entry, but it must still accept that completeness and descriptive depth are partly determined upstream by institutional workflows.

There is also a conceptual limitation worth stating explicitly: a national aggregator improves discoverability, but it does not automatically solve every issue of coverage. Some material may still be absent, delayed, restricted, or represented only indirectly. NordLit should therefore treat NVA as a highly valuable national entry point rather than as a perfect or exhaustive mirror of all Norwegian research output. Its greatest strength lies in providing a coherent, institutionally grounded route into Norwegian research metadata; its greatest weakness is that coherence at platform level cannot completely remove variation at record level.

2.1.2 Norwegian Journals and Publishers

Beyond the national infrastructure, NordLit also includes a substantial set of individually harvested Norwegian journals and academic publishers. These include open-access scholarly publishing platforms such as Septentrio at the Arctic University of Norway, the Bergen open-access environment, the open journal system at NTNU, the open journal system at OsloMet, journals hosted by the University of Agder, journals hosted by the University of Oslo, and academic publishers such as Universitetsforlaget, Novus Forlag, Orkana Forlag, and the electronic publishing environment at the Oslo Metropolitan University. Individual journals included in their own right include, for example, Polar Research, the Scandinavian Journal of Military Studies, Arctic Review on Law and Politics, the Nordic Journal of Information Literacy, and the Annals of Representation Theory, among others.

These journal-level sources matter because Norwegian scholarly publishing is not reducible to any single national aggregation point. Some journals publish outside the main national registration workflows, and others are important regional venues whose visibility in international databases is uneven. By including them individually, NordLit preserves the breadth of Norwegian scholarly publishing rather than relying only on a single upstream service.

2.2 Sweden

2.2.1 DiVA

What it is. DiVA, *Digitala Vetenskapliga Arkivet*, is both a shared search portal and a local publishing and repository system used by around 49 member organisations. The official DiVA pages describe it as Sweden's most used system for the registration and dissemination of scholarly publication data. It serves universities, university colleges, authorities, and research institutes, and it is designed not only to make publications available, but also to support local registration, open access, and institutional dissemination.

Why NordLit uses it. DiVA is one of the most important Swedish sources for NordLit because it combines substantial breadth with clear institutional provenance. It is particularly strong for dissertations, theses, reports, and institution-linked publications that are often unevenly represented in large international databases. For a regional search service, that combination of breadth and provenance is highly valuable.

DiVA is also useful because it is not merely a central index laid over many separate repositories. It functions as both a shared portal and a publishing environment. That means records are often well integrated into local institutional workflows, which can improve their stability and make their provenance easier to understand.

From the perspective of how NordLit collects the data, DiVA is a good example of a system that is shared and distributed at the same time. Rather than treating DiVA as one undifferentiated source, NordLit works through a registry of the participating organisations, gathering records organisation by organisation. In practical terms, this means that records from Lund University, the Royal Institute of Technology, Linköping University, Malmö University, Umeå University, and many others each come in with clear institutional provenance and can be filtered by the originating organisation if needed. At the time this manual was written, NordLit's coverage of DiVA numbers in the low millions of records.

What you can expect to find. Users should expect a wide range of Swedish scholarly outputs, including journal articles, dissertations, theses, reports, conference papers, and other publications linked back to the member institution where the work was registered or deposited. Since DiVA is both a shared portal and a local publishing environment, records often include stable identifiers and clear institutional attribution.

For NordLit users, one of the most valuable aspects of DiVA is that it can surface material that sits slightly outside the core of large international databases: doctoral dissertations, local report literature, university-published series, and other institutionally managed outputs. In a cross-Nordic context, this helps correct the tendency of international indexing systems to privilege only certain publication types and publication channels.

Advantages and limitations. The main advantage of DiVA is breadth across a large network of Swedish universities and research organisations while preserving institutional attribution. It gives NordLit a practical way to include a substantial part of Swedish repository and publication data through a shared infrastructure rather than through a long list of separate institutional systems.

The main limitation is variation in metadata richness across members and record types. Even within a shared platform, local registration practices still matter. NordLit can harmonise access to shared fields, but detailed search precision still depends on what each member institution supplies.

2.2.2 SwePub and Its Participating Repositories

What it is. SwePub is maintained by the National Library of Sweden and gives an overview of research produced in Sweden. The official service describes it as containing articles, conference contributions, and theses from Swedish universities and authorities, with contributions from more than 40 Swedish universities, colleges, and government agencies.

In NordLit, material from the SwePub family of sources is collected directly from the participating repositories. This currently includes the research information system at Chalmers University of Technology, the open archive at the University of Gothenburg, the publication database at Lund University, the student paper archive at Lund University, and the research publication archives of the Swedish University of Agricultural Sciences. Each is kept as its own identifiable sub-source but grouped together under a shared SwePub heading in the search interface.

Why NordLit uses it. For NordLit, SwePub is valuable because it provides a national aggregation perspective on Swedish institutional publication data. This can simplify discovery and support more coherent national coverage, especially when users need a Swedish overview rather than institution-by-institution searching.

Its value lies in national scale rather than in replacing local systems. Where a user would otherwise need to know which universities, colleges, or authorities are likely to hold relevant material, SwePub-style coverage offers a more central route into Swedish publication metadata. That makes it especially useful in cross-border work, where repeated local searching quickly becomes burdensome.

What you can expect to find. Users can expect broad Swedish metadata coverage drawn from universities, colleges, and authorities. In practice, this often provides an efficient route into Swedish publication data, including articles, conference material, and theses, but it still depends on the completeness and quality of the metadata supplied by upstream systems.

For NordLit, this makes the SwePub family of sources particularly helpful as a national overview. It can increase the chances of finding Swedish outputs that are less visible in global indexes, while still preserving a clear line back to the contributing institution. Student paper archives such as the one at Lund

University also make graduate-level work visible, which can be important in fields where student research contributes substantial empirical material.

Advantages and limitations. The main advantage is reach. Covering several major Swedish institutions through their own research information systems gives NordLit access to a national layer built from many contributors and therefore supports broad Swedish coverage without requiring institution-by-institution harvesting by the user.

The main limitation is that this coverage inherits upstream gaps and differences. NordLit can benefit from the national scale, but records remain only as detailed as the contributing systems make them. A national service can unify access more effectively than it can standardise all metadata in depth.

2.2.3 Other Swedish Sources

NordLit's Swedish coverage extends beyond the two large services above. It also includes the Linköping University electronic press, the e-book platform at Lund University, Stockholm University Press, Maritime Commons, several open-access journals hosted by Swedish universities such as those at the University of Gothenburg, Malmö University, Umeå University, Linnaeus University, and Lund University, and independent Swedish journals such as Barnboken, Iberoamericana, the Journal of Digital Social Research, the Journal of Intercultural Communication, the Journal of Intelligence Studies in Business, Rural Landscapes, Arkiv, and Socialmedicinsk tidskrift.

The pattern throughout is the same: each source contributes an identifiable slice of Swedish scholarly communication, and NordLit keeps them separately identifiable so that users can see where each record comes from.

2.3 Denmark

2.3.1 Forskningsportal

What it is. Research Portal Denmark is a national service for discovering Danish research. Its official documentation distinguishes between a global database and a local database. The local database is built from publication metadata harvested from Danish research institutions through the national DDF-MXD exchange format, and the portal documents a growing list of Danish data providers, including universities and other research organisations.

Why NordLit uses it. For NordLit, Research Portal Denmark offers a practical national route into Danish publication metadata. Rather than treating Denmark only as a set of separate institutional portals, NordLit can use an infrastructure that already works to gather, consolidate, and expose local research records. This is particularly helpful for cross-border searching, where repeated provider-by-provider searching quickly becomes inefficient.

The national exchange format is especially relevant here. Because the portal is already built around a structured mechanism for collecting publication metadata from Danish institutions, it reduces the amount of work that NordLit would otherwise need to repeat. That does not remove all differences in local practice, but it does create a clearer and more dependable national entry point.

In practical terms, Research Portal Denmark is the only source in NordLit that is acquired as a single national bulk release rather than through a protocol that delivers records one page at a time. The portal publishes a large compressed file containing all the metadata, and NordLit retrieves this file in whole. This is often the cleaner route when a national service already chooses to publish a comprehensive snapshot, because the snapshot is a well-defined version of what the country wished to make discoverable at that moment.

What you can expect to find. Users should expect broad Danish coverage, especially where local institutional records have already been harvested into the portal. The local database includes publication metadata from Danish data providers and covers a wide range of publication types associated with Danish research institutions. The portal also performs consolidation and matching work across records, which helps create a more usable overview than a purely provider-by-provider approach would provide.

For NordLit users, this is valuable because the portal can bring together Danish material that would otherwise remain distributed across many institutional systems and portals. It is therefore useful not only for recall, but also for making Danish research metadata easier to interpret in a national context.

Advantages and limitations. The main advantage is coherence. The portal already performs national harvesting and consolidation work that would otherwise have to be repeated elsewhere. This lowers integration complexity and makes Danish research metadata easier to approach as a national set rather than as a loose federation of local systems.

The main limitation is that the portal also includes a global-data layer. NordLit must therefore be careful to prioritise the parts of the service that genuinely strengthen Nordic discovery rather than simply reintroducing broad international coverage under a Danish label. As with other national aggregators, metadata quality also remains dependent on the quality of the local records supplied upstream.

2.3.2 Tidsskrift.dk

What it is. Tidsskrift.dk is the Royal Danish Library's national portal for professional, scholarly, and cultural journals in digital full text. The official service description states that it includes more than 200 journals and yearbooks and over 100,000 articles, many of them freely available. It also forms part of Denmark's National Strategy for Open Access.

Why NordLit uses it. Tidsskrift.dk is especially valuable because it surfaces Danish journals and yearbooks that are often central in Danish scholarly communication but less visible in international indexes. This is particularly relevant in the humanities, social sciences, education, history, law, and policy-related fields, where national-language journal literature can be essential.

For NordLit, the importance of such a source is clear. A regional discovery layer is weaker if it cannot adequately represent the journals through which domestic scholarly discussion actually takes place. Journal platforms such as Tidsskrift.dk therefore help correct the international bias that can arise when discovery depends too heavily on global citation indexes.

Tidsskrift.dk is also a good example of why NordLit's collection process needs to be adaptive rather than one-size-fits-all. The full historical depth of the platform is large, and its earliest date ranges are sometimes unstable. NordLit's approach therefore starts from recent years and works backwards, breaking difficult periods into smaller and smaller chunks when necessary, so that years which behave normally are not slowed down by years which do not. The aim is simply to make the collection reliable without missing usable material.

What you can expect to find. Users should expect Danish journal and yearbook material, often in full text, including both current and retrospectively digitised content. This can be especially useful when searching for

Danish-language research or practitioner-oriented literature that is influential nationally but underrepresented internationally.

Because the platform includes both newly published content and digitised older material, it is useful for both current evidence gathering and historical or contextual work. For some topics, especially in law, education, culture, and public debate, that retrospective depth can matter a great deal.

Advantages and limitations. The main advantage is stronger visibility for Danish-language and domestically oriented journal content. Tidsskrift.dk provides a direct route into a part of the Danish scholarly landscape that global systems often represent incompletely.

The main limitation is that metadata depth and consistency can vary across journals and time periods, especially in older retrospectively digitised material. A journal platform can greatly improve visibility without necessarily making every record equally rich for structured discovery.

2.3.3 Other Danish Sources

NordLit also includes several other Danish services, each filling a specific role. The institutional repository for organic research known as Organic Eprints contributes agricultural and environmental research literature. The open e-book platform at Aarhus University Library contributes scholarly books. The Danish preprint service hprints.org contributes pre-publication research material. The open journal system at Aalborg University and the open journal system at Copenhagen Business School contribute further journal content. The bulletin of the Geological Survey of Denmark and Greenland contributes earth-sciences literature. The European Journal of Economic Dynamics and Policy contributes economics research. The eprint archive of the Danish Research Centre for Magnetic Resonance contributes imaging-related research.

Together, these sources make Danish coverage in NordLit more than a two-service story. They fill gaps that neither the national portal nor the journal platform alone can address.

2.4 Iceland

2.4.1 OJS

What it is. Icelandic journal publishing is distributed across a small but important set of local platforms, including open journal environments associated with Icelandic institutions such as the University of Iceland. These platforms host scholarly journals and provide journal-level access to articles, editorial information, and publication workflows.

Why NordLit uses it. NordLit uses these platforms because Icelandic journal publishing is concentrated in relatively small domestic channels that are easy to overlook in larger international systems. An institutional journal platform can surface Icelandic-language and Iceland-focused articles that are central in local scholarly communication but much less visible in large external indexes.

This matters especially in a smaller research system. When national journal publishing is concentrated in a limited number of local venues, those venues become disproportionately important for fair regional discovery. If NordLit ignored them, Icelandic scholarship would be underrepresented not because it does not exist, but because its route to publication differs from that of larger systems.

What you can expect to find. Users should expect journal articles, often in open-access form, especially in fields where Icelandic-language publishing remains important. The platform is useful for finding domestically anchored scholarly literature with clear journal-level provenance and strong institutional context.

This can be especially valuable in subjects such as education, public administration, history, language, and policy, where local-language scholarship may be essential to understanding Icelandic research and debate.

Advantages and limitations. The main advantage is visibility for Icelandic journal literature that is not always indexed elsewhere. These platforms help NordLit include domestic scholarly communication that would otherwise be easy to miss.

The main limitation is scale. An institutional or journal-hosting platform does not aim to represent the entirety of Icelandic research output. It is therefore best treated as a focused complement to repositories and broader discovery services rather than as a complete national solution.

2.4.2 Skemman

What it is. Skemman is Iceland's shared digital repository. Its official description presents it as a repository for several Icelandic universities and institutions, including the University of Iceland, the University of Akureyri, Bifröst University, Hólar University, Reykjavík University, the Agricultural University of Iceland, the Iceland University of the Arts, and the National and University Library of Iceland. It contains student theses and, for earlier periods, research by academic staff.

Why NordLit uses it. Skemman is a key Icelandic source because theses and dissertations form an important part of the country's scholarly record. In a smaller research system, these outputs can contain valuable local evidence, methods, and case material that may never appear in journal form. Including Skemman helps NordLit avoid a journal-only view of Icelandic research.

This is particularly important when the research question concerns local implementation, educational practice, social conditions, or region-specific evidence. In such cases, thesis literature may be one of the richest sources available, even if it differs in status from formally published journal research.

What you can expect to find. Users should expect strong coverage of student theses and academic works from Icelandic universities, often with open-access files. This is particularly useful in areas where local context matters and where graduate work may contain substantial empirical material, literature reviews, or case-based analysis.

For NordLit, Skemman therefore adds a type of material that is often underrepresented elsewhere: locally grounded academic work that may be highly relevant to Nordic comparative or evidence-oriented searching even when it has not moved into commercial publishing channels.

Advantages and limitations. The main advantage is coverage of Icelandic academic work that may not otherwise be visible. In a small-country context, that contribution is significant because theses and dissertations often carry more local evidentiary weight than they would in a very large system.

The main limitation is that theses are heterogeneous. They vary in level, review status, and methodological maturity. NordLit can help users find them, but users must still assess them according to their purpose and standards of evidence.

2.4.3 Opin vísindi

What it is. Opin vísindi is an Icelandic open-access repository environment. Official descriptions present it as a repository for research materials and doctoral dissertations in open access on behalf of Icelandic universities and the National and University Library of Iceland, and University of Iceland materials describe it as a repository for open-access peer-reviewed articles and doctoral dissertations at Icelandic universities.

Why NordLit uses it. NordLit uses Opin vísindi because it complements Skemman. Where Skemman is especially strong for theses and student work, Opin vísindi helps extend Icelandic coverage toward deposited research publications and doctoral outputs. This makes it easier to represent Icelandic research in a form that is more comparable to repository material from the other Nordic countries.

In practice, that means NordLit can avoid treating Icelandic research as if it were visible only through student theses or local journals. Opin vísindi adds another route into research-level outputs that are openly deposited and institutionally grounded.

What you can expect to find. Users should expect open-access research outputs, deposited publications, and doctoral dissertations associated with Icelandic universities and research activity. In some cases, this may include article versions or other deposited research material that users need to find alongside Nordic material from the other countries.

For NordLit, the value of such a repository lies not only in access, but also in comparability. It helps Icelandic material sit more naturally beside repository-derived material from the rest of the region.

Advantages and limitations. The main advantage is better access to open Icelandic research outputs beyond student work alone. It strengthens the visibility of institutionally connected Icelandic research in a form that is easier to integrate into a regional search workflow.

The main limitation is that repository coverage depends on what institutions and authors choose or are able to deposit. It should therefore not be read as a complete mirror of all Icelandic research, but as an important open-access route into part of it.

2.4.4 Other Icelandic Sources

NordLit also includes the research archive of Landspítali, the Icelandic national university hospital, which is especially valuable for medical and health-services research, and the Arctic Portal Library, which provides a smaller but regionally focused collection of material connected to Arctic research.

2.5 Finland

2.5.1 Journal.fi

What it is. Journal.fi is the Finnish platform for scholarly journals operated by the Federation of Finnish Learned Societies. The Federation describes itself as the host of Journal.fi, and journals on the platform commonly state that they use the Federation's peer-review label where applicable. In practice, Journal.fi functions as a shared platform for a large part of Finland's open journal publishing landscape.

Why NordLit uses it. Journal.fi is important because Finnish journal literature, especially in Finnish and Swedish and in fields with strong domestic relevance, is not always indexed consistently by international databases. Including Journal.fi helps NordLit bring Finnish journal scholarship into a Nordic search workflow more reliably.

This matters especially in the humanities, social sciences, education, and regionally relevant policy fields, where local scholarly communication often takes place through learned-society journals and national-language publication channels rather than through the narrower set of outlets most visible internationally.

The Finnish platform hosts a large number of journals, and NordLit treats each participating journal as a separately identifiable stream. This means that coverage is broad without losing sight of the fact that the material is produced by many different editorial communities.

What you can expect to find. Users should expect a wide range of Finnish scholarly journals and society-based publications, many of them open access and many with clear information about peer review, editorial practice, and publication policies. The platform is especially useful for journal literature that is central to Finnish academic debate but less visible in international discovery systems.

For NordLit, Journal.fi therefore improves not only recall, but also balance. It helps make Finnish scholarship more visible on its own terms rather than only insofar as it has already been absorbed into international indexing infrastructures.

Advantages and limitations. The main advantage is stronger visibility for Finnish journal publishing that is locally important and often openly available. Journal.fi gives NordLit a direct route into a substantial part of Finnish scholarly journal communication.

The main limitation is that Journal.fi is a platform, not a complete national bibliography. It gives strong coverage of participating journals, but not all Finnish scholarly publishing passes through it. NordLit should therefore treat it as an important journal source rather than as a full account of Finnish research output.

2.5.2 Finna

What it is. Finna is Finland's national search service for materials from libraries, archives, museums, and other organisations. The official service describes Finna as bringing together cultural and scientific resources from hundreds of Finnish organisations, and its material-provider pages show that these include both memory institutions and research-relevant providers.

Why NordLit uses it. Finna is useful for NordLit when it provides access to research-relevant publications, reports, and repository material that would otherwise remain scattered across many institutions. It can be particularly helpful for Finnish-language reports, public-sector publications, and specialised collections that matter in evidence work but do not sit neatly inside journal or university-repository channels.

In this sense, Finna is not valuable because it is narrowly research-focused. It is valuable because, used carefully, it can expose parts of the Finnish knowledge landscape that would otherwise remain fragmented or overlooked.

Finna is also a particularly interesting case from the point of view of how the data are gathered. The service holds a very large volume of material, and there is a practical ceiling on how many records can be returned through any one request. NordLit therefore collects Finna one month at a time, covering the full period from the early twentieth century up to the present. If a particular month contains too many records to fit within the safe limit, that month is automatically split into smaller pieces. In plain terms: rather than trying to pour the whole river through a narrow tap, NordLit fills many smaller cups in sequence, keeping track of which cups are done and which still need to be filled. Because NordLit is looking specifically for research material, it also asks Finna to filter the results to textual formats such as books, journals, documents, theses, and other text material, so that the very broad cultural holdings of the service are narrowed to the portion most useful for research discovery.

What you can expect to find. Users should expect very broad coverage, including both research-oriented and non-research-oriented material. In NordLit, that breadth is useful only when handled selectively. The research-relevant parts include publication archives, institute publications, public-sector reports, and domain-specific collections from agencies and research organisations.

For example, a search concerned with policy, environment, transport, welfare, or economic governance may benefit from Finnish public-sector and institute publications that are discoverable through Finna but not always well represented in traditional scholarly indexes.

Advantages and limitations. The main advantage is reach across a large number of Finnish organisations. Finna can help NordLit surface material that is highly relevant to evidence work but does not belong neatly to journal-based or repository-based discovery alone.

The main limitation is noise. Because Finna is much broader than a research database, NordLit must use it selectively to preserve a research-focused experience. Without careful filtering, the breadth that makes Finna useful could also make it unwieldy.

2.5.3 Finnish Institutional Repositories

What they are. In addition to national-level discovery services such as Finna and shared infrastructures such as Theseus, Finland also has a set of important university and institutional repositories that expose research outputs directly from the institution responsible for them. These repositories are especially useful because they preserve clear provenance and often contain open-access copies, theses, dissertations, publication series, and self-archived versions of research articles.

For NordLit, this group matters as a distinct source family. It currently includes eighteen institutional repositories, covering universities and research organisations such as Aalto University, the Åbo Akademi research portal, the University of Lapland, the Finnish National Library, the University of Helsinki, the University of Oulu, the University of Jyväskylä, the Natural Resources Institute Finland, LUT University, the National Institute for Health and Welfare, the Finnish government publications platform, the University of Eastern Finland, the University of Vaasa, the University of the Arts Helsinki, Tampere University, and the University of Turku. While these services differ in local scope and metadata richness, they share a common function: they make institutionally grounded research outputs visible in a way that is often more direct than a national discovery layer can provide.

Helda deserves particular mention here. Helda is the University of Helsinki's open repository and is one of the strongest Finnish institutional sources for openly available scholarly material. It includes research publications, dissertations, theses, books, publication series, reports, and other material produced at the University of Helsinki. In practical terms, Helda is valuable not only because of the size of the University of Helsinki, but also because it provides a comparatively clear and stable route into a major part of Finland's university-based research output. NordLit collects Helda through a dedicated route as well as through the shared institutional harvesting described above.

Why NordLit uses them. NordLit uses Finnish institutional repositories because national services and broad discovery platforms do not always expose institutional research in the same way or with the same level of contextual clarity. An institutional repository often gives a cleaner line back to the university responsible for the work, and it may provide fuller local metadata, clearer series information, or direct access to an open version of the publication.

This is especially important for long-form and institutionally published material. Reports, dissertation series, working papers, departmental publications, and self-archived article versions are often easier to find and understand when they are reached through the institutional repository rather than only through a national aggregation layer. In that sense, these repositories strengthen NordLit not merely by adding records, but by improving provenance, traceability, and access.

Helda is a good example of this advantage. Because it is tied directly to the University of Helsinki's own repository and open-access workflows, it can surface University of Helsinki material with a degree of institutional specificity that would be harder to preserve if NordLit relied only on broader Finnish discovery systems. The same logic applies, in different ways, to each of the other Finnish institutional repositories in the set.

What you can expect to find. Users should expect a broad range of institution-linked research outputs, especially doctoral dissertations, master's theses, publication-series items, reports, books, book chapters, deposited articles, and other materials that universities make openly available through repository workflows. In some repositories, users may also find metadata for research data, teaching materials, or administrative publication series, although the research relevance of such material varies.

For NordLit, the main value of this source family is that it broadens discovery without detaching records from their institutional setting. A dissertation at the University of Helsinki, a self-archived article at the University of Turku, or a publication deposited at Aalto is easier to interpret when the repository context remains visible. The user can see not only the bibliographic record, but also the institutional environment in which the record was created, maintained, and exposed.

These repositories are also useful because they can reveal material that is less consistently represented in international indexes. University-published series, Finnish- or Swedish-language dissertations, deposited author manuscripts, and institution-specific report literature are all examples of outputs that may be highly relevant in Nordic evidence work while still remaining partially hidden in more general discovery systems.

Advantages and limitations. The main advantage of Finnish institutional repositories is provenance. They provide a direct connection between the record and the institution responsible for it. That improves traceability and often helps users understand whether they are looking at a journal article, a dissertation, a university series publication, a self-archived manuscript, or another kind of institutional output.

A second advantage is access. Institutional repositories frequently provide open versions of material that may otherwise be difficult to obtain quickly, especially dissertations, theses, reports, and self-archived article versions. For NordLit, this is practically important because discoverability is more useful when it is paired with a realistic path to the item itself.

The main limitation is fragmentation. Even when institutional repositories use similar repository software or metadata standards, they remain local systems with local practices. Metadata depth, language handling, type labels, and collection structure can therefore vary considerably from one institution to another. NordLit can harmonise the records after gathering, but it cannot remove all local differences at the source.

There is also some overlap between this source family and broader Finnish services. A record at the University of Helsinki or another institutional repository may also be discoverable through Finna or another aggregation layer. That overlap is not necessarily a problem. In a discovery service, overlap can improve recall and provenance, provided that duplication is handled carefully and the source relationship remains clear. For that reason, NordLit should treat Finnish institutional repositories as a complementary layer: they add institutional precision and open-access visibility to the broader Finnish discovery landscape rather than replacing the national services already described above.

2.5.4 Theseus

What it is. Theseus is the open repository of Finland's Universities of Applied Sciences and is provided by Arene, the Rectors' Conference of Finnish Universities of Applied Sciences. The official service describes it as giving online access to theses and publications from Finnish Universities of Applied Sciences.

Why NordLit uses it. Theseus is valuable because Universities of Applied Sciences produce a large amount of practice-oriented work in education, health, welfare services, technology, and public-sector development. This kind of material is often highly relevant for policy and implementation questions but less visible in international journal-based discovery systems.

For NordLit, that is an important strength. A regional discovery service should not represent Nordic research only through conventional university and journal channels. Applied and professionally oriented work can be essential, especially when the question concerns services, implementation, practice, or evaluation.

What you can expect to find. Users should expect theses and publications from the applied-sciences sector, often with full text. This includes work that is particularly useful when the research question concerns local implementation, service design, professional practice, organisational development, or applied evaluation.

In many cases, this material may be more immediately relevant to practitioners and policy users than a journal article would be. It therefore widens the evidentiary base that NordLit can expose.

Advantages and limitations. The main advantage is improved discovery of applied and professionally oriented Finnish work. Theseus helps bring a distinctive and practically important part of the Finnish higher-education landscape into Nordic discovery.

The main limitation is that these outputs vary in review status, methodological design, and intended audience. They can be highly useful, but they must be assessed carefully in relation to the user's purpose rather than treated as interchangeable with peer-reviewed journal literature.

2.5.5 Other Finnish Sources

Beyond the sources above, NordLit also includes Helsinki University Press, Tampere University Press, the VTT Research Information Portal of the VTT Technical Research Centre of Finland, the Åbo Akademi open journal system, the editorial platform Editori, Silva Fennica, and individual Finnish journals such as Estetika, Redescriptions, the Nordic Journal of African Studies, the Nordic Journal of Legal Studies, and the Nordic Journal of Migration Research. Each of these contributes to a fuller picture of Finnish scholarly publishing that neither the national search service nor the shared institutional repositories could provide alone.

2.6 Cross-National and Global Sources

2.6.1 OpenAlex

What it is. OpenAlex is an open bibliographic catalogue of the global research system. It connects works, authors, institutions, publication venues, publishers, funders, topics, and related entities in a single open knowledge graph. In NordLit, it is used as a supplementary discovery source rather than as a Nordic national infrastructure. Its value lies especially in its breadth and in the fact that many journals, publishers, repositories, and indexing sources contribute metadata that would otherwise be difficult to gather one by one.

How NordLit uses it. NordLit uses OpenAlex selectively. The purpose is not to import OpenAlex wholesale, because doing so would turn NordLit into a general international search service rather than a Nordic discovery layer. Instead, OpenAlex is used to support targeted coverage of selected journals, publishers, and source families that are relevant to the Nordic literature landscape and that are not always available through a stable national or institutional endpoint. In these cases, NordLit queries OpenAlex for records matching the intended journal or publisher, checks that the returned material belongs to the target source, and then harmonises those records into the same shared NordLit format used for other sources.

This approach is especially useful for smaller journals and open-access publication channels. Some of them do not expose a reliable harvesting interface of their own, but their articles are nevertheless represented in OpenAlex through publisher metadata, Crossref, repository metadata, or other upstream sources. OpenAlex therefore helps NordLit include relevant publication venues without requiring a bespoke collector for every small journal.

What you can expect to find. Users should expect OpenAlex-derived records in NordLit to include scholarly works such as journal articles, books and chapters, conference outputs, datasets, theses, preprints, and other research outputs when those works have been captured by OpenAlex and match NordLit's source-selection rules. For NordLit users, the most visible OpenAlex use case is selected journal and publisher coverage. A record may include title, authors, publication year, venue information, DOI or other identifiers, open-access information, citation-related metadata, and links to the source record where available.

The important point is that OpenAlex broadens discovery, but only within NordLit's defined scope. NordLit does not treat every OpenAlex record as relevant. It uses OpenAlex as a controlled input for specified sources, not as an unrestricted world database.

What it does not include, or does not guarantee. OpenAlex is broad, but it is not a fully curated Nordic bibliography. It is only as complete and accurate as the upstream metadata it receives and the matching rules that connect those records to journals, publishers, institutions, and works. Some records may lack abstracts, language information, affiliations, references, or full source detail. Publication types and language labels may also vary in precision. In addition, OpenAlex does not solve the problem of scholarly relevance by itself: a record can be present in OpenAlex and still fall outside a user's substantive research question.

For that reason, OpenAlex-derived records in NordLit should be interpreted as discovery records. They can be very useful for finding literature, especially from publication venues that are hard to harvest directly, but users should still verify the final bibliographic details and access conditions at the original source. This is consistent with the broader NordLit principle that the source record remains the point of authority.

2.7 How NordLit Uses These Sources

Taken together, these sources show why a Nordic discovery layer is useful. Some sources are national aggregators, such as SwePub and Research Portal Denmark. Some are shared institutional infrastructures, such as DiVA, Skemman, and Theseus. Some are broad national discovery systems, such as

Finna. Others are journal platforms, such as Tidsskrift.dk, Journal.fi, and Icelandic journal-hosting environments. Still others are individual journals, small academic publishers, or selected source families represented through OpenAlex. Each of them captures a different part of the Nordic research landscape. NordLit's role is to bring these strands together in a way that is easier to search, easier to document, and easier to understand.

This also means that NordLit must be selective. Not every source should be used in exactly the same way. Repository-first sources are often best for dissertations, theses, and institution-linked publications. Journal platforms are often best for domestic-language article literature. National aggregators are often best for broad coverage with clear provenance. A good Nordic discovery service therefore does not flatten all sources into one undifferentiated pool. It uses each source for what it does well while keeping provenance visible and preserving a clear route back to the original record.

3 How NordLit Works

The earlier chapters explain why NordLit is needed and which national sources matter. This chapter explains the basic logic of the system itself. NordLit is not a single act of collecting followed by a single search screen. It is a staged workflow in which the data are gathered, cleaned, standardised, indexed, and then presented through a web application. Each stage has a different function. Together they create a search service that is easier to use and easier to explain than the underlying source landscape would otherwise allow.

At a high level, the workflow is straightforward. Many small source-aware collectors gather descriptive information from repositories, catalogues, journal platforms, and national services. Their output is then passed through a harmonisation stage that extracts a shared minimum set of fields. After that, the many different ways sources describe publication types are translated into a single shared vocabulary. The result is loaded into a dedicated search engine. A web application on top of that search engine lets the user run searches, filter results, open the original source pages, and export records in useful formats.

What matters most is that these stages are separated rather than collapsed into one process. That separation makes the system easier to maintain and easier to understand. If a collection problem arises, it can be fixed in the collection layer without redesigning the web interface. If publication-type mapping needs improvement, that can be changed without rethinking how records are gathered. If the search interface evolves, the cleaned and standardised data can still remain stable underneath. The staged design therefore supports both transparency and maintainability.

3.1 From Source to Search

In practical terms, NordLit can be understood as a pipeline of five steps.

The first step is collection. Descriptive information is gathered from source systems through source-specific routines. These routines do not all work in the same way, because the source systems do not all expose their data in the same way. Some answer structured metadata requests page by page. Others provide a search-style interface. Others publish a single large download. In each case, the aim at this stage is simply to obtain a faithful local copy of what the source exposes, and to save it in a simple line-by-line structured format so that later stages always know what to expect.

The second step is harmonisation. Once records have been gathered, they still look very different from one source to another. Field names differ, nesting differs, and the same conceptual information may appear in different places. One source may expose authors as a structured list, another may put them into standard Dublin Core metadata fields, and a third may bury them inside a more elaborate structure. The harmonisation stage translates that variety into one shared shape. Every record, no matter where it came from, ends up with the same small set of clearly defined fields.

The third step is standardisation of publication types. Even after harmonisation, the publication type is still whatever the source called it. Three strings such as Article in journal, academic article, and journal-article all mean the same thing, but they look different. The standardisation step uses carefully maintained translation tables, one per source family, to convert these many variations into a single shared vocabulary. That is what makes it possible for a user to tick a single box such as Journal Article and have it behave consistently across every source in the system.

The fourth step is loading into a search engine. The standardised records are loaded into a dedicated search engine that can look through millions of records in a fraction of a second. Only selected fields are indexed for searching, and the right kind of text analysis is applied so that case and certain diacritics do not get in the way.

The fifth step is presentation. A small web application queries the search engine on the user's behalf, returns results, computes filter counts, and streams exports. The web page that the user actually sees is the final layer of the system; the serious integration work has already been done behind the scenes.

3.2 Why the Workflow Is Staged

A staged workflow matters because the underlying source landscape is heterogeneous. If NordLit tried to collect, clean, standardise, index, and present each source in one live process, the system would quickly become fragile. A small source-side change could break the entire user-facing service. By separating the stages, NordLit isolates different kinds of work from one another.

This also improves interpretability. A user or maintainer can ask different questions at different stages. Was the record gathered at all. Was it harmonised correctly. Was the publication type mapped reasonably. Was it loaded into the search engine. Does it appear in the interface. These questions are much easier to answer when the pipeline is explicit. A staged system is therefore not merely a technical convenience. It is part of making NordLit inspectable and trustworthy.

There is another advantage as well. Staged processing allows NordLit to keep the source-specific complexity where it belongs. Gathering from Norway's national research archive is not the same as gathering from Finland's national library service. Cleaning records from a library catalogue is not the same as cleaning records from a repository. The point of the pipeline is not to deny those differences. It is to absorb them in the background so that the search layer can still behave more consistently for the user.

3.3 What a NordLit Record Contains

At the core of the system is a shared record model. After harmonisation, each NordLit record carries a fixed set of fields that are used throughout the later stages. These include:

- a short identifier telling NordLit which source the record came from;
- a stable link back to the record on the original source website;
- the main title of the work;
- a list of author names;
- a single extracted publication year, or an empty value where no reliable year can be derived;
- an abstract or summary, or an empty string where none is available;
- the publication type exactly as labelled by the original source;
- the source's own internal identifier, kept for tracing and deduplication purposes.

After standardisation, two further fields are added. One is the standardised publication type, which supports filtering and more consistent interpretation across sources. The other is a brief note recording how that standardised type was derived. This is important because publication-type classification is not a purely mechanical truth. It is an interpretive layer added for practical discovery purposes, and keeping a note makes it inspectable.

A NordLit record is therefore not simply a copy of a source record. It is a transformed record whose purpose is discovery across sources. That transformation is intentionally limited. NordLit does not try to recreate every field from every source in the search layer. It preserves the fields that are most useful for search, filtering, provenance, and export, while still keeping a clear route back to the original source record.

3.4 Provenance, Traceability, and Duplicate Records

Because NordLit integrates records from multiple infrastructures, provenance matters. Users need to know where a record came from and how to reach the original landing page. The system therefore preserves source labels and source links wherever these can be constructed or extracted. This is one of the reasons NordLit remains a metadata service rather than a full-text redistribution service. The source remains visible throughout.

Traceability also matters for maintenance. The harmonised record retains the source's own identifier as an internal tracing field. This makes it possible to follow a record back through the processing pipeline and to understand whether a problem originates in gathering, harmonisation, or later stages.

At the same time, NordLit should not be described as a finished global deduplication system. The harmonisation stage removes duplicates within the output produced for a given source when the same source identifier has been recorded more than once, for example because a collection file has been appended across several runs. That is useful, but it is not the same as work-level deduplication across providers. The same publication may still appear more than once where it is exposed by more than one source. This is not a failure of the concept so much as a normal consequence of aggregating metadata from overlapping infrastructures.

3.5 What the User Sees and What the User Does Not See

Most of the complexity described above is intentionally hidden from ordinary users. They interact with a single search interface, not with a set of source-collection routines and cleaning jobs. That is appropriate. A discovery service should not force users to think like data engineers in order to perform a search.

Even so, it is useful for the manual to state clearly that the simple appearance of the web application rests on a layered architecture. Users search a prepared search index, not the live source systems directly. That is what makes the search interface fast enough to be practical and consistent enough to be documented.

What users do see are the parts of the system designed to support trust: stable source labels, links back to the original record, understandable filters, and exports that can be used outside the application. In other words, the complexity is hidden, but the provenance is not.

4 Gathering Nordic Metadata

Gathering is the first operational layer of NordLit. Its purpose is to collect descriptive information from the selected Nordic sources and preserve it locally in a structured form that can later be cleaned, standardised, and indexed. Because the Nordic source landscape is so varied, NordLit does not rely on a single universal collector. Instead, it uses a set of source-specific gathering routines that share a common overall philosophy while adapting to the technical characteristics of each source.

This is an important design choice. A system such as Norway's national research archive behaves very differently from a system such as Finland's national search service, which in turn behaves differently from a small journal platform. A reliable gathering layer therefore needs to be source-aware rather than pretending that all sources can be queried in exactly the same way.

4.1 General Principles

Across the whole gathering layer, the routines follow a consistent set of principles. They are resumable, so that long collection runs can continue after interruption rather than starting from the beginning. They keep a small progress file on disk so that they know where they left off. They are polite to the source servers, identifying themselves clearly, waiting for responses, and in many cases pausing deliberately between requests. They also aim to be robust, handling temporary errors, malformed responses, and unstable pagination patterns in source-specific ways.

These principles matter because the sources differ not only in structure but also in reliability. Some are large national infrastructures that can support substantial collection traffic when handled responsibly. Others are small institutional systems that need gentler pacing. A regional discovery service cannot simply treat every endpoint as interchangeable. Responsible collection is part of maintaining the source relationships on which the whole system depends.

4.2 How Collection Is Coordinated

NordLit's many source-specific routines are coordinated by a single orchestration layer. This means an administrator can initiate a full refresh with one command, and the system will then run as many routines as safely fits alongside each other, keeping each one's log separate so that problems can be diagnosed afterwards. For a quick test, the orchestrator supports a sample mode that gathers only a handful of records per source, which is useful for checking that everything is reachable without committing to a full run.

This arrangement balances efficiency with source-specific control. Each routine retains its own logic and its own state, but they can still be managed together as part of one larger refresh workflow. It also means that if one source is temporarily unavailable, only that one routine is affected. The others continue without interruption.

4.3 The Main Ways NordLit Obtains Records

Although there are many individual routines, they fall into a small number of families.

The most common approach is to ask the source for its records through a standard metadata-harvesting protocol widely used by academic repositories. In plain terms, the source offers a long, structured list of records and NordLit walks through that list page by page, keeping a bookmark so that a long run can be paused and resumed. This is the route used by most of the repository-based sources: the shared Swedish research archives, the Finnish institutional repositories, the University of Helsinki open repository, the Finnish Universities of Applied Sciences repository, the Icelandic shared repository, the Icelandic open-access repository, the Danish journal platform, the Finnish journal platform, and many smaller journal and platform sources.

A second approach is to use a source's own dedicated request interface. Norway's national research archive and Finland's national search service each offer a direct interface designed for searching their own holdings. NordLit's gathering routines for these sources are written to work specifically with those interfaces, because the shape of the data is distinctive.

A third approach is to retrieve a single prepared bulk file. Denmark's national research portal publishes all of its metadata as a compressed download, and NordLit retrieves the file as a whole, with support for resuming an interrupted download so that a slow connection does not force the entire file to be transferred again.

A fourth approach, used for a small number of journal platforms where the standard metadata-harvesting protocol is not available or reliable, is to read publicly available web pages directly. This is slower and more fragile, so it is used only where necessary.

A fifth approach is used for many individual journals and selected source families. OpenAlex provides a worldwide open catalogue of scholarly publications, and NordLit can query it for records whose journal title, source, or publisher matches a specified target. This is how many small single-journal sources can be covered, because it allows a journal to be added with relatively little source-specific code as long as the journal is reliably represented in OpenAlex and the returned records can be filtered back to the intended source.

4.4 Source-Specific Gathering Patterns

4.4.1 Norway's National Research Archive

Collection from Norway's national research archive is technically the most specialised part of the gathering layer. The archive's request interface has a known tendency: certain bookmark positions always fail, regardless of how often the request is retried. If a routine simply tried to work through the whole archive in one pass, it would get permanently stuck the first time it hit one of these bad bookmarks and lose the rest.

NordLit's approach is a careful four-phase strategy. A forward pass walks through records from oldest to newest, stopping when it meets a bad bookmark. A backward pass walks from newest to oldest, stopping when it meets the same kind of obstacle from the other direction. The gap in the middle is then filled by breaking it into one-hour windows and gathering each window separately. Windows that still fail are broken down into ten-minute pieces, and, if necessary, into one-minute pieces. Anything still failing at the one-minute level is logged as a known gap and skipped. Finally, all the collected pieces are merged into one consolidated, sorted file.

In everyday terms: rather than trying to read a whole book straight through when some pages are stuck together, NordLit reads from the front, from the back, and then fills in the middle by tackling smaller and smaller sections until what remains unreadable really is unreadable. This strategy is elaborate, but it is justified because Norway's national archive is a strategically important source and its request interface has known limits.

4.4.2 Shared Swedish Repositories

The Swedish shared repository environment is served through the standard metadata-harvesting protocol, but it is not one endpoint; it is a registry of many participating organisations. Rather than treating it as one undifferentiated source, NordLit's routine iterates through each member institution's dedicated endpoint in turn. This preserves institutional breadth while still treating the environment as one common technical family.

This matters because the environment is both shared and distributed. The gathering logic needs to understand both facts at once. NordLit therefore benefits from the environment's broad institutional network without losing the organisation-level context that helps explain provenance and coverage.

4.4.3 Finland's National Search Service

Finland's national search service is very large and has a practical ceiling on how many records can be returned for any single query. Rather than trying to pull everything through one enormous request, NordLit splits the collection into monthly windows covering the period from the early twentieth century up to the present. Before each month is gathered, the system first asks the service how many records exist in that window. If the count is above a safe limit, the month is automatically split into smaller sub-windows, and those are gathered separately.

In plain terms: NordLit uses many small, well-defined requests instead of one large unreliable one. This makes the whole collection reliable, resumable, and predictable even when the source is enormous. It also means that incremental maintenance is easier. The collection can be understood as a set of manageable pieces rather than as one opaque run.

4.4.4 Denmark's National Research Portal

Denmark's national research portal is handled differently again. Instead of walking a live request interface page by page, NordLit retrieves a single prepared bulk file published by the portal itself. Downloads can resume from where they stopped if the connection is interrupted, and the file itself is a clear snapshot of what the portal chose to expose at that moment.

When a reliable bulk file exists, it is often better to use that than to simulate a bulk collection through repeated page-sized requests. This reduces complexity and often improves reproducibility, because the file is a single well-defined artefact.

4.4.5 Many Smaller Journals via OpenAlex

A large share of the individual journals included in NordLit are covered by asking OpenAlex to return records whose journal title, source, or publisher matches a specified target. OpenAlex answers through a predictable request interface, and NordLit walks the response from one record to the next while preserving source identifiers and links where they are available.

This approach is important because it keeps the per-journal code very small. Adding a new journal whose metadata is represented in OpenAlex typically requires only a short configuration file stating the journal's name and matching rule. The shared gathering routine then handles the details: paging through the response, filtering out results that do not actually belong to that journal, avoiding duplicate identifiers, remembering progress, and writing the output in NordLit's preferred shape.

OpenAlex is not used here as an unrestricted global database. It is used as a practical route to selected publication venues that matter for Nordic discovery but do not always provide a dependable harvesting endpoint of their own. This distinction is important for interpreting coverage: the presence of OpenAlex-derived material in NordLit should be read as targeted source coverage, not as a promise that all OpenAlex records are included.

4.4.6 Other Source-Specific Patterns

Several other sources have their own quirks. The Danish journal platform works through the standard metadata-harvesting protocol but uses a date-slicing strategy, starting from recent years and working backwards, with adaptive window shrinking when difficulties appear. The Finnish journal platform supports a discovery mode in which it first asks the platform which journals exist and then gathers each one separately. The bundle of Finnish institutional repositories uses the standard metadata-harvesting protocol but tries several candidate addresses per institution, because different repositories expose their endpoints under slightly different paths. Small Icelandic and Finnish journals auto-detect their own endpoints from a homepage by trying a list of common locations. A small number of journals are collected by reading public web pages directly, because no reliable metadata-harvesting interface is available.

Taken together, these examples show that NordLit's gathering layer is not one generic routine with a few options. It is a family of source-aware tools built around a common philosophy.

4.5 Progress Files and Failure Recovery

One of the most important practical features of the gathering layer is resumability. Collection runs are long, and source systems are not perfectly stable. For that reason, every routine writes a small progress file to disk recording, for example, the bookmark or token it is currently using, which pages have been completed, how many records have been written, and which identifiers have already been seen.

This matters for both efficiency and trustworthiness. Without resumability, a failure near the end of a long run could force the entire source to be collected again. With resumability, NordLit can continue from a meaningful checkpoint. This makes the update process more realistic and less wasteful.

Failure recovery also improves documentation. When the manual explains that NordLit is built for transparent and responsible use, that principle applies internally as well. A gathering layer that records its progress and its failures is easier to inspect than one that merely succeeds or disappears.

4.6 What Gathering Produces

The output of gathering is not yet the user-facing NordLit record. Each source writes a source-specific structured file under its own folder. These files preserve much more of the source-specific structure than the final search index does. In many cases they still contain raw source-formatted descriptions, technical context, and other information useful for later cleaning and debugging.

This is deliberate. The gathered file is a preservation layer, not the final search layer. By keeping the gathered files distinct from the cleaned and indexed records, NordLit makes it possible to return to the source material when something in the later stages needs to be checked or improved.

5 Cleaning and Harmonising Records

After gathering, NordLit still has a collection of source-specific files rather than one coherent discovery dataset. The purpose of the harmonisation stage is to transform those heterogeneous files into one common minimal record structure suitable for later standardisation, indexing, and display.

In effect, this is the stage where NordLit stops thinking mainly in terms of source protocol and starts thinking in terms of searchable records.

5.1 Why Harmonisation Is Necessary

The need for harmonisation is straightforward. Even when two sources both use the same underlying metadata protocol, they do not necessarily expose the same fields in the same way. One source may place the title in a straightforward Dublin Core field. Another may rely on a more complicated structure or a source-specific extraction. A third may encode rich metadata that still needs interpretation. The same problem applies to authors, dates, abstracts, identifiers, and publication types.

A search service cannot work well if every source continues to speak its own internal language. Harmonisation is the stage that translates those differences into one shared vocabulary at the field level. It does not erase the source. It prepares the source for cross-source discovery.

5.2 The Shared Record Shape

The harmonised output uses a deliberately small and practical structure. Each harmonised record contains the source identifier, the best available link back to the source, the main title, a list of author names, a single extracted publication year or an empty value where no reliable year can be derived, an abstract string or an empty string, the publication type as labelled by the source, and the source's own internal identifier.

This shape is intentionally limited. NordLit does not attempt to force every possible metadata field from every source into the search layer. The aim is to preserve the fields that are most important for search, filtering, interpretation, export, and traceability. A smaller shared shape is easier to maintain and easier to explain.

5.3 How the Core Fields Are Extracted

The cleaning logic is source-specific, but its aim is common. Titles are taken from the source's most authoritative title field, with fallback logic where needed. Author names are extracted from whichever structure each source uses, which might be an explicit author list, standard creator or contributor fields, or a more elaborate structure that has to be read carefully.

Years are handled conservatively. A source might express the year as *2023*, as *2023-06-15*, as *2023/06/15*, or embed it inside a longer string. A shared helper looks for any four-digit number that could plausibly be a year (anything from the year 1000 onwards and covering the current century) and extracts it as an integer. If nothing plausible can be found, the record's year is left empty, meaning simply *not available*. This matters because it is more honest to leave a field empty than to fabricate a year from ambiguous data.

Abstracts are taken from summary or description fields where available. Publication types are retained as the source presents them, even though they will later be standardised separately.

5.4 Reconstructing Useful Links

Source links require similar care. They are not always given as a straightforward field. For some repositories the link has to be reconstructed from a known pattern using the source's internal identifier. For example, a repository identifier such as *oai:skemman.is:1946/711* is not itself a working web address, but the numeric part can be combined with the repository's public URL pattern to produce *https://skemman.is/handle/1946/711*, which does work. NordLit performs these reconstructions wherever a reliable pattern is known.

This matters in practice because users clicking on a result in NordLit rely on the link to take them to the real source page. A result without a working link is much less useful than a result with one. The reconstruction logic ensures that, wherever possible, every record points somewhere real.

5.5 Source-Specific Harmonisation

The harmonisation stage is not merely a set of mechanical renamings. For each source, it includes source-aware rules about where the useful fields actually reside. Records from Norway's national archive are read from the API's own nested structure for titles, contributors, dates, abstracts, and types. Records from shared Swedish repositories may rely on already-extracted fields but can also fall back to parsing the underlying structured metadata when necessary. Records from the large family of standard-protocol repositories, including the University of Helsinki repository, the Finnish Universities of Applied Sciences repository, the shared Icelandic repository, the Icelandic open-access repository, the Danish journal platform, and many others, rely substantially on standard Dublin Core fields. Records from Finland's national search service use its own structures, including format labels and summary fields. Records from Denmark's national research portal use the fields found in the downloaded bulk file.

This source awareness is one of the reasons a general-purpose metadata aggregator can miss important details. NordLit's harmonisation layer is explicit about those differences. It does not assume that a title field means the same thing everywhere. It asks, source by source, where the most useful title for search actually is.

5.6 Deleted Records, Empty Titles, and Skipped Records

Not every gathered record should become a searchable NordLit record. Some sources expose deleted records as part of normal harvesting. These are useful for maintenance and provenance, but they are not useful as visible search results. The harmonisation stage therefore skips deleted records where appropriate.

Records without a usable title are also excluded at this stage. NordLit is a discovery service, and a record without a recoverable title is difficult to present meaningfully in the interface and difficult to use responsibly in export.

Similarly, records that cannot be harmonised due to malformed or unusable source content may be skipped.

This means that the harmonised dataset is not a raw mirror of the gathered material. It is an operational discovery layer built from that material.

5.7 Before and After: A Harmonisation Example

Consider a source record from a Swedish shared repository. In its raw form it might look something like this:

- an identifier, for instance *oai:DiVA.org:su-12345*;
- a set of extracted fields containing a title (*Nordic welfare policies after 2008*), a list of author names, a publication year string (*2019*), an abstract, and a raw publication type (*Article in journal*);
- a deleted flag set to false;
- a source URL pointing to the record on the Stockholm University repository.

After harmonisation, the same information becomes one simple, flat record:

- source: the shared Swedish repositories identifier;
- source link: the Stockholm University repository URL;
- title: *Nordic welfare policies after 2008*;
- authors: a list with two names;
- year: 2019 (now an integer);
- abstract: the text as given;
- publication type: *Article in journal*;
- internal identifier: *oai:DiVA.org:su-12345*.

In plain language, the same information is now in a flat, predictable shape. The deep nested structure has gone. The year string has become a number. The identifier is preserved for tracing purposes. A record like this can be filtered, searched, and exported consistently regardless of which source it came from.

5.8 Duplicate Handling and Its Limits

The harmonisation stage performs deduplication by internal identifier within each harmonised output. This is mainly intended to correct cases where collection files have accumulated duplicate records across multiple runs. In other words, it protects against accidental technical duplication inside one source's pipeline.

This should not be confused with global deduplication across the whole NordLit corpus. If the same publication appears in two different source systems, and therefore has two different source identifiers, both records may remain. That is an important limitation to document clearly. NordLit improves discovery across infrastructures, but it does not yet claim to collapse every cross-source duplicate into a single canonical work record.

5.9 What Harmonisation Produces

The result of harmonisation is a set of cleaned files, one for each source or source group. These are the first genuinely cross-source-comparable layer of the NordLit pipeline. They can already be inspected directly and are substantially easier to reason about than the raw gathered files.

Even so, they are not yet the final records used by the web application. One important layer still remains: publication-type standardisation. That layer deserves its own chapter because publication types are one of the most variable and interpretively difficult parts of Nordic metadata integration.

6 How Publication Types Are Derived

Publication type is one of the most useful fields in a literature search system, but it is also one of the least uniform across sources. Different infrastructures classify outputs differently. Some distinguish very carefully between article subtypes, monographs, reports, theses, and conference outputs. Others use broad categories such as *text*, *publication*, or *article*. Some rely on human-readable local labels. Others expose machine-oriented semantic codes. A shared Nordic search service therefore cannot simply display source types as if they were directly comparable.

NordLit addresses this through a separate standardisation stage. This stage preserves the source's original type label while adding a new field that maps the local label into a shared cross-source vocabulary. It also records a brief note explaining how the interpretation was made, so that the reasoning remains inspectable rather than invisible.

6.1 Why Standardisation Is Needed

Users often want to narrow a search by broad output form. They may want journal articles only, or reports and policy documents, or doctoral dissertations, or books and chapters. Without standardisation, this is difficult across sources because each source speaks its own type language.

Standardisation therefore serves a practical purpose. It creates a common filtering layer. At the same time, it also serves a transparency purpose. A controlled vocabulary makes it easier to explain how publication types were interpreted and where uncertainty remains.

What standardisation does not do is magically remove all ambiguity. Sometimes a source label is too broad to classify with confidence. In those cases NordLit uses categories such as *unknown*, *missing*, or *other* rather than pretending to know more than the source really allows.

6.2 How the Translation Tables Were Developed

The translation tables are static and hand-crafted rather than generated through live automated guessing. This is important because publication-type mapping should be inspectable, repeatable, and easy to revise. NordLit therefore works from observed source values and explicit source-specific interpretation rules. The tables were built by examining the actual publication-type labels that appeared in the gathered data, one source at a time, and deciding for each one which standard category it should map to.

In practice, this means that the tables are large. They cover thousands of different source-specific strings across all the Nordic sources currently included. A single source may contribute dozens of distinct labels, each one mapped to the standardised vocabulary by a single, explicit rule. That is unglamorous work, but it is also what makes the final filter behaviour trustworthy.

6.3 The Shared Vocabulary

The standardised vocabulary used in NordLit includes around thirty-five categories. Among them are the main scholarly article types (journal article, review article, editorial, letter or comment, short note or communication, registered report), the conference output types (conference paper, conference abstract, conference poster, conference proceeding), the book-related types (book, edited book, book chapter), report-style outputs (report, working paper, policy brief), the various levels of theses (doctoral dissertation, licentiate thesis, master's thesis, bachelor's thesis, and unspecified student thesis), other research outputs (preprint, dataset, software, patent), reference-style material (encyclopedia entry, reference entry), and non-academic outputs such as magazine articles, newspaper articles, popular science articles, professional articles, media contributions, artistic outputs, and presentations or lectures. In addition, three special values exist: *other*, *unknown*, and *missing*.

This vocabulary is intentionally pragmatic. It is broad enough to support meaningful filtering, but it is not so fine-grained that it simply reproduces the fragmentation of the source labels. In a search service, usability matters. The vocabulary is therefore designed for discovery rather than for modelling every bibliographic nuance of every source.

6.4 Special Values: Missing, Unknown, and Other

The special values deserve explicit explanation. *Missing* means that the source record did not contain a publication type at all, or that the relevant field was empty. *Unknown* means that a type string was present but could not be mapped with sufficient confidence. *Other* is used where a record clearly describes some real output type, but one that falls outside the controlled vocabulary NordLit currently uses.

These values are not defects to be hidden. They are part of responsible classification. A trustworthy search service distinguishes between lack of information, ambiguous information, and uncommon but real output types. Users can even filter by these values directly, for example to inspect which records have no usable publication type at all.

6.5 A Layered Approach to Mapping

The standardisation logic proceeds in layers. First, it looks up the source-specific translation table for the source of the record. If the raw type string is found there, that translation is used directly. If not, and the raw type looks like a structured machine code (for example, ending in */article* or */doctoralThesis*), the last segment of that code is extracted and looked up. If that also fails, two broader fallback tables are tried: one covering common article-type terms from the worldwide registry of scholarly publications, and one covering common types from the standard metadata-harvesting protocol.

If all of the above fail, a final layer of keyword matching is applied. If the raw string contains words such as *doctoral*, *phd*, or *väitöskirja* (the Finnish term for doctoral dissertation), the record is classified as a doctoral dissertation. If it contains *master*, *pro gradu*, or *diplomityö* (Finnish terms for master's-level work), it is classified as a master's thesis. If it contains *conference paper*, the record is classified as a conference paper. And so on for journals, books, reports, preprints, datasets, software, patents, and artistic outputs.

In plain terms: even if a type string is unusual, as long as it contains a recognised keyword in English or in a Nordic language, the record can still be classified. The Finnish term for a doctoral dissertation, for example, is recognised directly. So is the Finnish term for a master's thesis. This makes the standardisation robust against unfamiliar sources and languages without needing a separate translation entry for every possible variation.

6.6 How Publication Types Are Derived by Source

Different sources benefit from the layered approach in different ways.

Records from Norway's national research archive expose relatively detailed local labels such as *academic article*, *academic literature review*, *non-fiction chapter*, *report research*, *degree PhD*, or *artistic performance*. These map comparatively cleanly onto the shared vocabulary and provide some of the highest-confidence classifications in the system.

Records from shared Swedish repositories rely mainly on standard semantic codes that can be reduced to their meaningful endpoints, such as *article*, *doctoralThesis*, *masterThesis*, or *conferencePaper*. In addition, each repository also uses a handful of short internal code letters that have their own translations. NordLit maps both the semantic codes and the short codes.

Records from the family of repositories including the University of Helsinki, the Finnish Universities of Applied Sciences, the shared Icelandic repository, the Icelandic open-access repository, the Danish journal platform, and the Finnish journal platform often expose the same standard semantic codes. Where the semantic code is missing, they may fall back to human-readable Finnish, Swedish, or Icelandic terms that the keyword layer can still interpret.

Records from the Finnish journal platform also use a useful fallback based on set membership. Records belonging to an article set are treated as journal articles for discovery purposes. Records belonging to a news or notice set are handled more generically.

Records from Finland's national search service expose format labels such as *Kirja* (book), *Artikkeli* (article), or *Väitöskirja* (doctoral dissertation). NordLit prefers the translated human-readable label where the service supplies one, and otherwise falls back to the raw format value.

Records from Denmark's national research portal use the portal's own type labels, which are then mapped through a dedicated translation table.

Records from the many individual journals gathered through the worldwide registry of scholarly publications use that registry's own type codes. Most of these are simple: *journal-article*, *book-chapter*, *proceedings-article*, and so on. These map cleanly onto the shared vocabulary.

6.7 What the Standardisation Stage Produces

The output of publication-type standardisation is a set of records in which each harmonised record has gained two additional fields: a standardised category drawn from the shared vocabulary, and a brief note recording how that category was reached (for example, *direct match on 'AcademicArticle'* or *keyword match (doctoral) in 'Doctoral dissertation, monograph'*).

The distinction between original and standardised type remains important. NordLit does not discard the original source type. It adds an interpretive layer on top of it. That design choice supports both usability and inspectability. A user can filter on the standardised category while still being able to inspect what the source originally said.

7 Indexing for Search

Once records have been cleaned and their publication types have been standardised, they are ready to be loaded into the search engine. Loading is the stage that turns a processed metadata collection into a responsive search service. Without it, NordLit would still be a useful metadata-processing pipeline, but it would not yet be an interactive discovery tool. The point of the search engine is to make the prepared records searchable at scale and to support the filtering and export behaviours used by the web application.

7.1 Why a Dedicated Search Engine Is Used

A dedicated search engine is used because NordLit needs fast full-text-like querying over selected metadata fields, predictable filtering, filter-count calculations, and large-scale export. A simple relational database could store the records, but it would not provide the same search-oriented behaviour as naturally. A dedicated engine is designed for exactly this kind of retrieval layer.

This does not mean that NordLit performs arbitrary full-text search across every piece of content it has seen. The indexed fields remain selected metadata fields. The search layer is powerful, but it remains a metadata search layer.

7.2 The Indexed Fields

Records are loaded with a clearly defined set of fields. These include the title, authors, publication year, abstract, source link, source label, the original publication type, the standardised publication type, the source's own internal identifier, and the publication-type mapping note. Not all of these fields are meant to be searched directly in the same way. Some are indexed mainly for filtering or traceability. Others, such as the source link and mapping note, are preserved without being used as search text.

This distinction is important. Search systems work best when indexed fields have clear roles. NordLit therefore separates fields used for matching from fields used mainly for display, filtering, or provenance.

The current search layer indexes titles and abstracts in two complementary ways. The default text fields are analysed so that case and many diacritical marks do not block matching. Additional accent-sensitive subfields are also created for users who explicitly want special characters to matter. This makes it possible for the interface to offer both broad, forgiving searching and stricter special-character searching without changing the underlying record model.

7.3 Text Analysis and Matching

The search engine applies standard text analysis to the main text fields: the text is broken into word-sized pieces (tokenisation), letters are made lowercase (lowercasing), and certain accented or diacritical characters are reduced to their plain equivalents (ASCII folding). In practical terms, the default search mode is case-insensitive and tolerant of many diacritic differences. A user searching for *leikskoli*, for example, can match records containing *leikskóli*. This default behaviour is intentional, because users often do not know in advance which diacritics appear in a Nordic-language title or abstract.

NordLit also includes an exact special-character search option. When this option is enabled, the query is sent to accent-sensitive fields. In that mode, *leikskoli* and *leikskóli* are no longer treated as equivalent. The option is useful when a user needs literal character matching, for example when distinguishing between terms that differ only by a diacritic or when testing whether a source actually uses a particular spelling.

When a search is performed, the query text is matched against the title and abstract. The default behaviour for multiple search words is that all of them must match somewhere in those fields unless the user changes the logic with Boolean operators. NordLit does not currently search every field in the record body and it does not search the full text of publications. It searches the fields most central to bibliographic discovery: title and abstract.

NordLit does not automatically expand British and American English spellings. A search for *color* should not be assumed to retrieve *colour*, and a search for *behaviour* should not be assumed to retrieve *behavior*. Users need to handle British and American English as separate variants when that distinction matters for their search.

7.4 Sorting and Retrieval

Results are sorted first by publication year in descending order, with records lacking a usable year placed after dated records, and then by how well they match the search query. This means that the interface tends to prioritise newer material while still using textual relevance within that structure.

That choice reflects a practical compromise. Users often expect recent literature to appear early, but they also expect the query to matter. Sorting by year and then by match quality is therefore not a claim that the newest record is always the most relevant. It is simply the retrieval policy currently implemented in the application.

7.5 Filter Counts and Grouped Filters

The web application also needs count information so that users can narrow the result set. The back end therefore calculates how many records match per source group, how many match per publication type within each source group, and how many match per year.

Source filtering is grouped rather than presented only as a flat list. This is important because several underlying source identifiers belong naturally to one broader source family, for instance the different shared Swedish repositories or multiple Finnish institutional repositories. Grouping makes the interface easier to understand while still preserving the underlying identifiers needed for actual filtering. Countries are ordered consistently (Norway, Sweden, Denmark, Finland, Iceland), as are the sources and journals within each country.

The current interface uses cross-filtered facet counts. This means that each filter panel is counted against the other active restrictions, but not against its own selected values. If a user selects a database or publication type, the year counts update to show the years available within that source/type selection, while the database and publication-type counts remain interpretable in relation to the current search. Conversely, if a user selects a year, the database and publication-type counts update, while the year list remains readable as a year facet. This avoids the common problem in which a selected facet immediately counts itself down and becomes harder to compare with neighbouring options.

7.6 Loading and Refreshing the Search Engine

The records are loaded into the search engine in batches after each refresh of the pipeline. Because NordLit is not a live pass-through service to the source systems, it depends on periodic refreshes of a prepared search layer. The current schedule for those refreshes is not fixed by the scripts themselves and can be decided operationally.

8 The Web Application

The NordLit web application is the user-facing part of the system. It is implemented as a lightweight web server connected to a single web page. The application does not search the source systems directly. Instead, it queries the prepared search engine and renders results, filters, and exports for the user.

This distinction is essential. The application feels like one search service because the difficult integration work has already been done behind the scenes. The interface is therefore simple by design, not because the underlying source landscape is simple, but because the earlier stages of the pipeline have already absorbed much of that complexity.

8.1 General Design

The application consists of a small server back end, a single main web page, and a small amount of browser-side code that manages search rows, filters, result rendering, selections, exports, and saving or loading of the search state. The back end provides three endpoints: one to deliver the main page, one to handle searches, and one to handle exports.

This architecture is appropriately lightweight for NordLit's purpose. The system does not need a large front-end framework in order to provide search, filtering, and export. The important requirement is clarity and reliability, not visual complexity.

8.2 The Layout

When the page loads, the user sees three main regions. At the top is a dark header with the NordLit logo and an *Import/Export Search* menu on the right. On the left is a sidebar with filter panels. In the centre is the main search area with result cards below it. The header stays fixed at the top of the screen even when the user scrolls, and the sidebar scrolls independently from the main area so that filters remain visible while the user pages through results.

Above the search box, the interface shows a quiet build label, for example *Testing build · v0.0.0 · Updated 1 Apr 2026*. This is not meant to compete with the search controls. It is a small version marker so that testers and maintainers can tell which interface build they are using when reporting behaviour.

The search area also includes a compact legend explaining the main query syntax. The legend is deliberately placed close to the search box because it supports query construction at the moment the user needs it. Nearby helper text reminds users that British and American English must be handled as separate variants.

8.3 What the Search Box Actually Searches

The central search function operates over titles and abstracts. This should be stated very clearly in the manual because users often assume that a discovery service searches everything. NordLit does not search full text. Nor does it currently search every metadata field. It searches the fields judged most useful for bibliographic retrieval.

This has practical implications. A publication with a very sparse title and no abstract will be harder to retrieve by topical searching than a record with a rich abstract. That is not a defect of the interface. It is a direct consequence of the metadata available for search.

8.4 Boolean Search Construction

The interface supports Boolean logic. The first search row acts as the main query line. Users can type expressions such as *"school attendance" AND intervention*, combining words, phrases, and the standard operators *AND*, *OR*, *NOT*, with parentheses for grouping and quotation marks for phrases. As the user types, a syntax-highlighting layer colours the operators: *AND*, *OR*, *NOT*, parentheses, quotation marks, truncation marks, and proximity operators each receive their own visual treatment. The aim is not decoration. It helps users see whether the query has been written as intended.

Additional rows can be added by clicking *Add row*. Each new row has its own Boolean operator drop-down choice (AND, OR, or NOT) and its own text area. When the search is run, the front end builds a single combined query string and sends that to the back end. For example, a first row reading *"school* *attendance”* and a second row set to *OR* and reading *truancy* becomes the combined query *”school attendance” OR (truancy)*.

This design matters because NordLit is intended for systematic and semi-systematic searching, not only for quick keyword lookup. The ability to construct a query across several rows makes the interface more approachable than asking all users to type a complex Boolean statement in one line, while still preserving the resulting Boolean logic. Pressing Enter in any row runs the search. A *Clear* button resets the full interface to its initial state: search rows, text, filters, selections, paging, and result display are cleared together.

8.5 Proximity, Wildcards, and Special Characters

The search box supports truncation and wildcard searching with an asterisk. A query such as *migrat** can retrieve records containing terms such as *migration*, *migrating*, or *migratory*, depending on what is present in the indexed title and abstract fields. The same wildcard symbol can also be used inside quoted phrases. For example, *“teacher qual*”* can retrieve phrase-like matches such as *teacher quality* or *teacher qualification*. More complex phrase patterns, such as *“teach* quality”* or *“teacher qual* policy”*, are also supported.

Because unrestricted wildcard expansion can be computationally expensive, NordLit applies a safety limit to wildcard expansion in quoted phrase queries. This prevents broad patterns from overloading the search engine. In ordinary use this should not affect well-formed searches such as *“higher educ*”* or *“student achiev*”*, but very broad patterns, especially those beginning with an asterisk, may behave less exhaustively or more slowly. Users should treat leading wildcards as a specialist tool rather than as a normal first choice.

The *NEAR/#* operator supports proximity searching. A query such as *student NEAR/2 performance* asks for the two terms to occur within two words of one another. The number can be changed: *NEAR/3*, *NEAR/5*, and similar expressions request wider windows. Proximity matching is useful when a user wants a relationship between words without insisting on an exact phrase. In the current implementation, proximity matching is unordered, so it can retrieve both *student … performance* and *performance … student* within the specified distance.

The special-character option changes whether accented and diacritical characters are treated broadly or exactly. By default, special characters are folded to their plain equivalents, so *leikskoli* can match *leikskóli*. When exact special-character search is enabled, the same query becomes stricter: *leikskoli* no longer matches *leikskóli*. This option should be used when literal spelling matters more than broad recall.

8.6 Automatic Correction of Quotation Marks

One of the most practical user-facing features is the automatic correction of quotation marks. Before the query is submitted, the interface replaces curved or typographic quotation marks (those produced by word processors and other formatted-text environments) with ordinary straight quotation marks. Both double and single typographic quotation marks are corrected.

This feature deserves explicit mention because it solves a real and common problem. Users often paste Boolean searches from word processor documents, PDFs, websites, or email, and those environments frequently replace straight quotes with typographic ones. Many search systems interpret typographic quotes badly or not at all in phrase searching. NordLit therefore corrects them automatically so that phrase-based Boolean search behaves more reliably.

NordLit also checks for common query problems before returning results. If a search string is missing a closing quotation mark, missing a parenthesis, ends with an operator such as *AND*, or uses a malformed proximity operator, the interface shows a specific explanation rather than a generic loading error. This is meant to help users repair the query quickly. A search service that supports Boolean logic should not merely fail when the syntax is wrong; it should tell the user what kind of repair is needed.

8.7 Filters by Source and Publication Type

The left-hand panel allows users to narrow results by source group and by publication type within each group. This grouping is important because the underlying source identifiers are not always the most meaningful way to present the dataset to ordinary users. A user usually wants to think in terms such as *Norway's national research archive*, *the shared Swedish repositories*, *Finland's national search service*, or *the Finnish journal platform* rather than in terms of every internal identifier used by the pipeline.

Publication-type filtering is nested under source groups. This reflects an important design principle. Publication-type interpretation is source-aware. A standardised type such as *journal article* is used across the system, but the interface still lets the user see and apply that filter in relation to a specific source group. In practical terms: a user can say *journal articles from the Norwegian national archive* in one click, without having to first select the source and then the type separately.

A combined *Journal Platforms* group gathers the many individual journals harvested through journal platforms or OpenAlex-supported source matching into one expandable group. Within this group, the sub-items are the individual journals themselves rather than publication types, and they are ordered by country.

The source filter also includes small information icons next to source markers. Hovering over, or focusing on, one of these icons opens a brief source description. These descriptions are intended to help users decide whether a database, repository, journal platform, or journal website is relevant for a particular search before they apply a filter. The tooltips do not replace this manual, but they make the most important source-level guidance available directly in the interface.

This design supports transparency. The interface does not merely offer one giant type list detached from provenance. It keeps type filtering close to the source context from which the records come. As the user selects and deselects filters, the filter counts update in relation to the other active facet category, so that source/type selections affect the year counts and year selections affect the source/type counts.

8.8 Year Filtering

The application supports both year-range filtering and exact-year selection. At the top of the year panel are two input boxes labelled *From* and *To*, and below them a list of exact years with their counts. The two can be combined. A user can specify a range such as 2020 to 2023 and also tick individual earlier years of interest (for example, for a particular event year). The system treats the two as alternatives joined by *OR*: any record that matches either the range or any of the exact years is kept.

The year list also contains special categories. *Other* is used for records with known years that fall outside the ordinary year list used by the interface, including 2027–2030 and 2033–2099. *Missing* is used for records where no year is available. Separating these two cases matters because a record with a future or out-of-band year is different from a record that lacks a usable year altogether.

This provides a useful combination of flexibility and precision. It is also important to state that year extraction depends on the metadata available in the record. If no reliable year can be derived during harmonisation, the record appears under *Missing* rather than behaving like a dated record.

8.9 Result Display

Each result card displays the title, the year if one is available, the source label, the publication-type label if one is available, the authors on one line, the abstract if available, and a clickable link to the source URL when one exists. Any leftover formatting instructions from the source are removed before display so that results read as normal text rather than as raw markup.

Search terms in the title and abstract are highlighted with a yellow background. The system extracts meaningful words from the query (ignoring Boolean operators and very short words) and highlights their matches in the displayed text. This helps the user see at a glance why each record was returned.

The results area header shows the current visible range and total, for example *Showing 1 to 25 of 3,412 results (137 pages)*. Below the results is a pager with previous and next buttons and numbered pages. Two pages to either side of the current one are always visible, along with the first and last pages. Missing spans are shown as .... The page size can be changed using the *Show* drop-down in the toolbar, with available sizes of 10, 25, 50, 100, and 250 results per page. The default is 25.

This design keeps the interface readable while still preserving the fields most users need for quick judgement. It also reinforces provenance, because the title links directly to the original source record rather than encouraging the user to treat NordLit as the final destination.

8.10 Selection

Each result card has a checkbox. When the user ticks one or more checkboxes, a selection bar appears at the top of the results area showing the current count of selected records, a small export button, and a clear button. The selection survives paging. Moving to a different page, adjusting the search, or changing filters does not lose what the user has already selected. This is important because it allows an iterative workflow: search, select a few good records, refine the query, select a few more, and continue until the user has a curated set.

8.11 Export of All Results and Selected Results

NordLit supports two export modes. Users can export the entire result set matching the current query and filters, or they can select individual records and export only those selected items. This is an important distinction. In some workflows the user needs the full search result set for documentation or screening. In others, they may wish to curate a smaller subset before export.

Supported export formats are:

- *CSV*: a simple spreadsheet file with one row per record;
- *Excel*: a formatted spreadsheet with a dark-blue header row and auto-sized columns;
- *RIS*: a plain-text format recognised by most reference managers including EndNote, Mendeley, and Zotero;
- *BibTeX*: a format used in LaTeX documents and by Zotero;
- *JSONL*: one structured record per line, useful for programmatic analysis.

These are not merely convenience options. They support different downstream uses. The spreadsheet formats are useful for inspection and screening workflows. The reference-manager formats support citation management and reference import. The programmatic format is useful when a user or maintainer wants the most direct machine-readable representation of the indexed records.

The export behaviour is handled on the back end. When the user exports all results, the system reruns the underlying query and streams every matching record into the download without trying to load them all into memory at once. This lets NordLit export large result sets reliably. When the user exports selected records, the export is driven by the identifiers of those selected results. In both cases, the downloaded file is named in a way that includes the current date and the number of records, for example *Nordlit_20_04_2026_347_records.ris*, so that users can keep track of which export corresponds to which search.

The BibTeX and RIS exports pick the right kind of entry based on the publication type. Anything marked as a thesis or dissertation becomes a thesis entry. Anything marked as a book becomes a book entry. Anything marked as a conference output becomes a conference entry. Anything marked as a report becomes a report entry. Everything else becomes an article entry. This means that when the file is opened in a reference manager, each entry arrives with roughly the right icon and the right field layout.

8.12 Save and Load a Search

The header contains an *Import/Export Search* menu. The export option bundles the user's current query rows and operators, source selections, source-specific publication-type combinations, year range, exact years, page size, exact special-character setting, and a timestamp into a small file and downloads it. The import option reads a previously saved file and restores the search state.

This feature is particularly valuable for transparent research practice. It makes it easier to preserve not only what was found, but how it was searched. For collaborative or iterative searching, the ability to save and restore a search state is more than a convenience. It is part of making search decisions portable and inspectable.

A concrete use case: a literature reviewer documenting a search in a methods section can note that the search was exported on a particular date. Anyone else importing the same file in NordLit later sees the same combined query string, the same source selections, the same type filters, the same year filters, and the same relevant search-mode settings. The saved file lives on the user's own computer, not on a shared server, which keeps the workflow simple and avoids requiring user accounts.

8.13 A Typical User Journey

A typical NordLit search proceeds as follows. The user opens the page in a browser, notes the testing-build information if relevant, and types one or more Boolean search terms, possibly over several rows. They may use quotation marks for phrases, an asterisk for truncation or wildcard searching, *NEAR/#* for proximity, and the exact special-character option if literal diacritics matter. The interface corrects problematic quotation marks, validates the query, constructs the final query string, and sends that query to the back end. The back end searches the title and abstract fields in the prepared search engine. The application then shows the matching records and the available source and year filter counts.

The user can narrow the result set by selecting sources and publication types, adjusting the year range, or ticking specific years. As the user works with filters, the facet counts update across panels rather than simply counting down inside the selected panel. The user can inspect source tooltips when deciding which database, repository, or journal platform to include. They can page through the results, open records of interest in a new browser tab, and tick checkboxes to build a selection. When they have what they need, they click *Export selected* or *Export all* and pick the appropriate format. The file downloads with a date-stamped name. Optionally, the user also exports the search state through *Import/Export Search* so that they can reproduce or share it later.

If the user wants to start again, the *Clear* button returns the application to its opening state. This matters because a serious search session often involves testing several alternative strategies. A reliable reset is part of making the interface predictable.

This journey is simple on the surface, but it rests on the full staged architecture described earlier. The manual should therefore help users see both aspects at once: NordLit is easy to use because a great deal of source-specific work has already been done before the search begins.

9 Export, Documentation, and Reproducibility

One of NordLit's central practical goals is to support transparent literature searching rather than only ad hoc discovery. For that reason, export is not a minor convenience at the edge of the system. It is part of the core workflow. A search service is much more useful for evidence work when the result set and the search state can be taken out of the system and documented clearly.

9.1 Why Export Matters

In structured searching, finding records is only one stage of the work. Those records often need to be screened, imported into reference managers, reviewed by collaborators, archived with search documentation, or re-analysed later. If the export layer is poor, the search remains difficult to reuse even when the interface itself works well.

NordLit addresses this by providing several export formats rather than one. This makes the system adaptable to different downstream practices. It also fits the wider goal of avoiding unnecessary manual rework in the research process.

9.2 What the Export Formats Are For

The spreadsheet formats (comma-separated values and Excel) are useful when the next stage is inspection, sorting, annotation, or simple screening. They make the result set legible in general-purpose tools and are often the easiest formats for teams who are not working entirely inside citation software.

The reference-manager formats (RIS and BibTeX) are useful when the result set needs to move into citation managers or more traditional academic reference workflows. These exports are necessarily simplified representations of the underlying NordLit record, but they make the system usable within familiar bibliographic environments.

The programmatic format (one structured record per line) is the most direct machine-readable export. It is useful when the goal is technical inspection, pipeline reuse, or downstream processing in scripts. For some advanced users, this may be the most faithful export format because it stays closest to the full indexed record structure.

9.3 Saved Searches as Method Documentation

The saved-search file is especially important for reproducibility. It captures not only the textual query, but also the way the interface state was configured. That includes the presence of multiple Boolean rows, source restrictions, source-specific type restrictions, year filters, and page-size settings.

This means that NordLit supports two distinct but complementary forms of documentation. One is record export, which preserves what was found. The other is search-state export, which preserves how the search was configured. Together they make the search process easier to explain and to repeat.

A methods section can therefore say, plainly: *The search was carried out in NordLit on the given date using the configuration saved in the accompanying file. Running the saved file in NordLit reproduces the same combined query and filter state.* That is a substantially stronger documentation claim than simply naming the databases searched.

9.4 Practical Cautions

Even a strong export layer has limits. Exports are only as informative as the underlying metadata allow. A sparse source record will remain sparse after export. Similarly, citation-oriented formats require simplification and cannot preserve every internal detail of the NordLit record. Abstracts in the BibTeX format, for example, are truncated to a reasonable length so that the file remains manageable.

Users should therefore understand export as a strength of NordLit, not as a guarantee that every downstream tool will interpret every record perfectly. The original source remains important for verification when exact bibliographic detail matters.

Search documentation should also record interpretive choices that affect retrieval. Examples include whether exact special-character search was enabled, how British and American English variants were handled, and whether broad wildcard or proximity expressions were used. The saved-search file captures the interface state, but users should still describe the search strategy in ordinary language when preparing a formal report or review.

10 Limits, Trade-offs, and Interpretation

A manual that explains how NordLit works should also explain where interpretation is still required. NordLit makes Nordic metadata more searchable and more coherent, but it does not remove every difficulty inherent in the source landscape.

10.1 Metadata Quality Still Varies

NordLit can harmonise fields, but it cannot create metadata that the source never supplied. Some records have rich abstracts, stable identifiers, and clear publication types. Others are much thinner. This affects search behaviour directly. A richer record is generally easier to find and easier to interpret than a sparse one.

This is not unique to NordLit. It is a normal consequence of working with aggregated metadata. Even so, the manual should make the point clearly so that users do not mistake interface consistency for source uniformity.

10.2 NordLit Is Not a Full-Text Search Service

The application searches titles and abstracts in the prepared search engine. It does not search the full text of the underlying publications. For some users this may be perfectly appropriate. For others it means that NordLit should be understood as a discovery layer that points onward rather than as a replacement for all other forms of searching.

The practical implication is simple. A source record with no abstract and a very generic title may still be important, but it may be harder to retrieve by topic than a better-described record. A careful user therefore does not rely on NordLit alone to establish the absence of material. They use it to establish presence, and they verify the completeness of their picture against other sources where appropriate.

10.3 Duplicates May Still Exist Across Sources

NordLit reduces technical duplication within each source's harmonised output, but it does not yet perform comprehensive work-level deduplication across the entire multi-source corpus. Users should therefore expect that the same publication may sometimes appear more than once when more than one source exposes it.

This is especially likely in an environment where national aggregators, institutional repositories, and library systems overlap. The existence of such overlap is not, by itself, evidence of poor design. It is part of the reality NordLit is trying to make more manageable.

10.4 Publication-Type Mapping Is Useful but Not Absolute

Publication-type standardisation makes filtering much more useful, but it remains an interpretive layer rather than an infallible truth. Some source types are broad, local, or ambiguous. In such cases NordLit maps conservatively, and sometimes the result is *other* or *unknown*.

This is a strength rather than a weakness when described honestly. A trustworthy system should be cautious where the metadata are genuinely uncertain. The original publication-type label is always preserved alongside the standardised one, so a user can always check what the source itself said.

10.5 The Search Interface Reflects Current Implementation Choices

The search behaviour described in this manual reflects the current implementation. Fields, relative weights, sorting logic, filter-count behaviour, and interface features can change as the system develops. A manual should therefore distinguish between core principles and current implementation details.

The core principles are stable: shared Nordic metadata discovery, visible provenance, transparent export, and staged processing. Specific interface details, however, may evolve as NordLit is refined.

10.6 Users Should Still Verify at the Source

NordLit is designed to improve discovery, not to displace the source of record. A responsible workflow therefore still involves checking the original source record when exact bibliographic, licensing, repository, or access information matters. The title link on every result card is not an optional extra. It is part of the intended use of the system.

This is especially important when users are preparing formal evidence syntheses, reports, or publications. NordLit should make the route to the source easier and clearer, but it should not be mistaken for the final authority on every record.

10.7 A Pragmatic Conclusion

The best way to understand NordLit is as a practical regional infrastructure. It does not solve every problem of Nordic metadata integration, but it solves several important ones. It reduces repeated searching across many interfaces, improves the visibility of Nordic research, supports more structured filtering and export, and makes search workflows easier to document. It does this by being explicit about sources, explicit about processing stages, and careful not to promise more than the metadata can support.

That is already a substantial contribution. A good discovery system does not need to be perfect in order to be valuable. It needs to be understandable, useful, and honest about its own limits. NordLit is strongest when it is described in exactly those terms.